

JUSTIA

Williams v. State

Full Name: Williams v. State

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251 Ga. 749 (1983)

WILLIAMS v. THE STATE.

39641.

Supreme Court of Georgia.

Decided December 5, 1983.

Rehearing Denied January 18, 1984.

Lynn H. Whatley, John Thomas Chason, for appellant.

Lewis R. Slaton, District Attorney, Joseph J. Drolet, Assistant District Attorney, Michael J. Bowers, Attorney General, Mary Beth Westmoreland, Assistant Attorney General, for appellee.

Don C. Keenan, William J. Berg, amicus curiae.

BELL, Justice.

On February 27, 1982, the appellant, Wayne Bertram Williams, was found guilty of the murders of Jimmy Ray Payne and Nathaniel Cater and received two consecutive life

sentences. Appellant's motion for new trial was denied on December 16, 1982, and he appeals. We affirm.

1). Initially, we note that the state introduced evidence of ten other alleged murders to aid in establishing appellant's identity as the perpetrator of the murders of Payne and Cater. See Division 4, *infra*. Appellant's first enumeration of error concerns the state's use of circumstantial evidence based on expert comparisons of textile fibers found in the appellant's environment with fibers discovered on the twelve victims' bodies. The state's experts testified that they believed certain of the fibers which were associated with the victims matched others which had been recovered from appellant's home and cars he had used, and they opined that these matches established an inference that Williams had been in contact with the victims before *750 their deaths. This opinion was relied upon by the prosecution to support the further inference that Williams had killed the victims. Appellant now claims that the state failed to adequately demonstrate the scientific reliability of the fiber methodology which was employed by its experts, and that admission of the evidence based on these principles and techniques was therefore error.

It is for the trial court to determine whether a given scientific principle or technique is competent evidence. *Harper v. State*, 249 Ga. 519 (1) (292 SE2d 389) (1982). "The trial court may make this determination from evidence presented to it at trial by the parties; in this regard expert testimony may be of value. Or the trial court may base its determination on exhibits, treatises, or the rationale of cases in other jurisdictions." *Id.* at 525. We find no error here.

2). In his second enumeration of error Wayne Williams complains that he was denied his due process right to have experts of his choosing examine fiber evidence in the possession of the state crime laboratory. *Sabel v. State*, 248 Ga. 10 (6) (282 SE2d 61) (1981). In *Sabel* we held that a "criminal defendant on trial for his liberty is entitled on motion timely made to have an expert of his choosing, bound by appropriate safeguards imposed by the court, examine critical evidence whose nature is subject to varying expert opinion." *Id.* at 17-18. Appellant claims that his due process rights as recognized in *Sabel* were violated by a November 12, 1981 order of the court which gave his fiber expert, a California criminalist named Charles Morton, entry to the state crime laboratory and which permitted him to examine the fibers pertaining to Cater and Payne. Williams now contends that the scope of the order was too narrow, since Morton should have been allowed to examine fibers associated with the ten uncharged offenses used during the trial to establish plan, scheme, *modus operandi*, and identity. See Division 4, *infra*. Moreover, Williams contends that other restrictions imposed by the court's order operated to deny his due process rights by

preventing him from knowing the scientific instruments used by the state and the results it achieved using those instruments, thereby precluding him from developing an effective defense based on challenges to the methodology and conclusions of the state's fiber experts.

In his fifth enumeration of error, appellant further asserts that Morton was denied meaningful access to critical evidence in the Payne and Cater cases because, from time to time, the evidence was not available to him, and because he was forced to use equipment of the state crime lab which was not in good working order.

Although there is no dispute that the evidence in question was critical and that its nature was subject to varying expert opinion, we find that, under all the circumstances of this case, there is no merit in *751 these enumerations. We first deal with the question of which portions of the record appellant may rely upon to support his arguments. In addition to the pre-trial and trial record and transcripts, both Williams and the state direct our attention to an affidavit by Morton which was filed in support of appellant's motion for new trial. Despite the fact that Morton performed the greater part of the defense analysis of fiber evidence, he was never called as a witness, and the proffer of his affidavit constituted appellant's sole attempt to introduce Morton's firsthand account of his contacts with the crime lab and the obstacles its personnel allegedly placed in his path. However, the record shows that the court specifically refused to admit the affidavit as evidence; this refusal is separately enumerated as error by appellant, but, as we find in Division 19, *infra*, that exclusion was not erroneous. Hence, we will not consider Morton's affidavit for the purpose of determining the instant enumerations.

We begin our analysis by emphasizing that "the defendant does not have an absolute, unqualified right to examine such evidence. The motion for an independent examination must be timely made. The trial court in the exercise of its inherent power to conduct the proceedings before it, should impose appropriate safeguards to insure that the evidence is unchanged and preserved for evidentiary use at the trial. This would generally require that the defendant's expert be allowed to examine the substance in the state laboratory under the control and supervision of the state rather than relinquishing custody and possession of the substance to him. The request must be reasonable. If any valid reason exists for not permitting the defendant to use the substance for an independent analysis, the trial court may, as a matter of discretion, refuse to permit such an examination." *Patterson v. State*, 238 Ga. 204, 206 (232 SE2d 233) (1977).

a). We do not think that the court erred by failing to expressly permit inspection of fibers relating to the ten extrinsic offenses. The trial court's duty to mandate independent

inspection of critical evidence in the possession of the state does not arise unless defense counsel moves in a timely fashion and notifies the court of which critical evidence it seeks to inspect. If the defendant knows or in the exercise of reasonable diligence should know of the existence of potentially critical evidence in the possession of the state, yet fails to move in a timely fashion for its examination, then he or she cannot claim unfair surprise by its introduction at trial.

In the instant case it is clear that Williams cannot claim he was unfairly surprised by the introduction of the evidence in question, since the pre-trial record and transcripts are replete with information which shows that he clearly anticipated that evidence of this type was *752 in the state's possession and might be used by it at trial. Yet, despite the fact of his awareness that the prosecution might use evidence of extrinsic offenses, and despite the fact that the trial began December 28, it was not until February 1 that he apprised the court that he was dissatisfied with the court's omission to authorize his expert to inspect that evidence. Moreover, even if we assume that the court erred by failing to direct the state crime lab personnel to allow appellant's experts to examine the evidence relating to extrinsic offenses, the error was harmless. On at least two occasions ÔÇö January 18 and February 17 ÔÇö defense experts were in fact offered the opportunity to look at the fibers in question, but declined to do so.

b). Neither do we find that the trial court erred in its November 12 order by preventing Williams from examining the notes of the state's experts, their reports, and the physical products and results of their tests and experiments. Williams argues that, because the evidence in this case was so complex and voluminous, his constitutional right to an adequate opportunity to prepare his defense was not sufficiently served by affording him the bare opportunity to perform his own tests of critical evidence. Instead, he contends, fundamental fairness dictates that he also should have been given a meaningful opportunity to verify both the validity of the procedures used and the results reached by the state in order to be able to effectively rebut them at trial. However, the record shows that Williams never challenged the validity of any of the rules set out in the November 12 order, and, in particular, never raised the instant issue. Thus, for the reasons stated in subdivision (a), *supra*, we will not consider this issue on appeal.

c). We now consider appellant's argument, found in his fifth enumeration of error, that the prosecution and the state crime lab personnel violated his Sabel rights by failing to provide his expert with meaningful access to critical evidence. He contends that Morton was frustrated at each stage of the trial by being forced to use crime lab equipment which was

not in good working order and by being told that evidence he wanted to work on was not available. These contentions have no merit.

First, the allegation concerning the condition of the equipment Morton had to use was never brought to the trial court's attention, and, since, as already noted, the trial court properly excluded the Morton affidavit from the record, there is nothing to support the allegation on appeal. Next, in regard to appellant's allegations concerning the nonavailability of evidence, the record shows that only one such allegation was raised in the court below. At a January 20 and 21 hearing, defense attorney Binder claimed that the state was withholding fiber evidence pertaining to Payne and Cater, but the *753 state unequivocally denied that it had impeded or blocked Morton in any way or that evidence had been withheld. The court did not find that the state was in any way culpable, but, in an abundance of caution, ordered the state to give Morton another chance to look at the evidence in question. Mr. Binder indicated that such action would satisfy his objection. When it later became known that Morton had subsequently left for California without taking advantage of the opportunity provided by the court, the judge, upon Binder's motion, directed the state to open the crime lab for another defense expert, Bresee, who then apparently performed as much of an independent examination as the defense wanted at that point. These facts clearly show that the court adopted all the corrective measures which were requested by defense counsel, and therefore there is nothing to review on appeal.

3). It is undisputed that on August 14, 1981 Williams made a proper and timely request for production of scientific reports under OCGA 17-7-211 (a) (Code Ann. 27-1303), and in his third and fourth enumerations of error, Williams alleges that numerous violations of that statute occurred during the course of the trial.

a). One of appellant's complaints is that the trial court erroneously admitted the testimony of several state's experts concerning the results of scientific experiments, as no scientific report had been prepared from these experiments and provided to the defense. However, we have recently decided this issue adversely to appellant, *Law v. State*, 251 Ga. 525 (2) (307 SE2d 904) (1983),^[1] and this contention is without merit.

b). Alternatively, Williams argues that the experts' handwritten notes of their experiments should be construed as "scientific reports" for purposes of OCGA 17-7-211 (Code Ann. 27-1303). We disagree, and find that the Court of Appeals has correctly held that the notes and work products of a state's expert do not fall within the purview of OCGA 17-7-211 (Code Ann. 27-1303). *Sears v. State*, 161 Ga. App. 515, 516 (288 SE2d 757) (1982);

Hartline v. State, 161 Ga. App. 847, 849 (2) (288 SE2d 902) (1982); Hartley v. State, 159 Ga. App. 157, 159 (2) (e) (282 SE2d 684) (1981).

c). In another attempt to invoke the strictures of OCGA 17-7-211 (Code Ann. 27-1303), Williams argues that graphs *754 produced by an instrument known as a microspectrophotometer (MSP) constitute "scientific reports" and should have been delivered to him pursuant to his request under that statute. According to Larry Peterson, a microanalyst with the Georgia State Crime Laboratory, an MSP is an instrument designed to measure the color of microscopic materials such as fibers. Peterson testified that the instrument functions as follows: the material to be examined is first placed on a microscopic slide and put under a microscope's lens, and light is then passed through the material, on through the microscope, and into a spectrophotometer, which is perched on top of the microscope. Once inside the spectrophotometer, the light passes into a monochromator, an instrument which basically splits the light into its component parts or wave lengths. That information is then amplified, sent through a microprocessor, and recorded on a graph, which consists of a series of curves representing the color of the material. Dr Peterson used an MSP to compare the colors of fibers taken from the Wayne Williams environment to similar fibers found on some of the victims. For instance, state's exhibit #650 is a graph containing three curves, one depicting the dyes or colors present in a green fiber recovered from the hair of Nathaniel Cater, one depicting the dyes or colors present in a known green fiber removed from a green carpet in appellant's bedroom, and one depicting the dyes or colors of a green fiber removed from that carpet which had been tested for fading by being immersed for three days in Chattahoochee river water.

Williams argues that such a graph embodies the final test results and conclusions regarding the comparisons of fiber colors and is therefore a scientific report subject to discovery under OCGA 17-7-211 (Code Ann. 27-1303). Thirteen such graphs were introduced at trial over the defense objection that the state had not provided them to the defense pursuant to its request under OCGA 17-7-211 (Code Ann. 27-1303). The trial court overruled this objection, finding that the graphs did not compose "scientific reports." We agree. In State v. Madigan, 249 Ga. 571, 574, fn. 2 (292 SE2d 406) (1982), we briefly discussed the issue of what elements comprise a scientific report. We noted that the element common to the examples of scientific reports listed in OCGA 17-7-211 (Code Ann. 27-1303) was "that each includes the examiner's findings based on scientific analysis or his or her opinion." After examining Dr. Peterson's testimony, it is clear that the graphs appellant sought to exclude did not contain the conclusions of Dr. Peterson, but instead had to be interpreted by him in order to attain significance. For this reason, we

conclude that the graphs were not "scientific reports" and were not discoverable under OCGA 17-7-211 (Code *755 Ann. 27-1303).

4). In his sixth enumeration of error, Williams raises, inter alia, the issue of the admissibility of evidence of ten homicides which were not the subject of this trial.

We have held that evidence of other criminal acts of the defendant may be admitted if it "is substantially relevant for some purpose other than to show a probability that (the defendant) committed the crime on trial because he is a man of criminal character...." Walraven v. State, 250 Ga. 401, 407 (297 SE2d 278) (1982) (quoting McCormick on Evidence, 190 at 447, 2d Ed., 1972). The purposes for which evidence of extrinsic offenses may be offered include motive; intent; absence of mistake or accident (each is aspect of intent), plan or scheme (of which the crime on trial is a part); and identity. Walraven v. State, supra, at 408. Kilgore v. State, 251 Ga. 291 (3) (a) (305 SE2d 82) (1983); McCormick, supra, 190, at 448-451. To render evidence of extrinsic offenses admissible for any of these purposes, the state must show that the defendant was the perpetrator of the extrinsic offenses, and that there is a sufficient similarity or connection between the extrinsic offense and the offense charged, such that proof of the former tends to prove the latter. Walraven v. State, supra, at 408; Kilgore v. State, supra, at 296; Hamilton v. State, 239 Ga. 72, 73 (235 SE2d 515) (1977). Therefore, the issue before us is whether there is sufficient evidence tying Williams to the perpetration of the independent crimes and whether the independent crimes were similar or logically connected to the offenses charged so that proof of the former tends to prove the latter. A determination of this issue depends on an analysis of the facts surrounding the two charged crimes and the ten uncharged crimes.

Part of the evidence used by the state to connect appellant with these crimes consisted of comparisons of fibers and hairs discovered on the victims' clothing and bodies, and on items used in the recovery of their bodies, with fibers and hairs removed from appellant, his home and cars, and his German Shepherd dog. These comparisons were conducted by three state's experts: FBI Agent Harold Deadman, GBI employee Larry Peterson, and Royal Canadian Mounted Police employee Barry Gaudette.

To compare the hairs and fibers found in the appellant's environment with those found on the victims, the state's experts used a variety of microscopes. In this regard, Agent Deadman, a microanalyst, described as follows the microscopes which can be used to compare fibers: 1) a stereobinocular microscope, which can magnify a single fiber about seventy times, and which is used to visually compare fibers; 2) a compound microscope, which can magnify a single fiber approximately 400 to 500 times, and which, like *756 the

stereobinocular microscope, is used to visually compare fibers; 3) a comparison microscope, which can magnify two fibers side by side, and which is used to compare the microscopic and optical properties of the two fibers; 4) a microspectrophotometer (see Division 3 (c) of the opinion); 5) a polarizing light microscope, which is used to examine the optical properties of fibers in a more discriminating fashion than that provided by a comparison microscope; and 6) a fluorescence microscope, which is used to determine the type of light a fiber emits after it has been illuminated with a certain type of light. Additionally, a scanning electron microscope was used occasionally by the state's experts. From the transcript it is difficult to discern which of these microscopes were used to conduct each particular fiber comparison; however, Larry Peterson implied that the stereobinocular microscope, the comparison microscope, and the polarizing light microscope were used to conduct all of the comparisons. It is also apparent that several of the other microscopes, in particular the microspectrophotometer and the fluorescence microscope, were used to compare some selected fibers.

The types of fibers and hairs which Agent Deadman testified were taken from appellant and his environment, along with the items from which they were taken, are as follows: 1) violet acetate and green cotton fibers representing the composition of a bedspread found in appellant's bedroom; 2) green and yellow nylon fibers used to fabricate the carpet found in appellant's bedroom; 3) dog hairs removed from appellant's German Shepherd; 4) yellow rayon and acrylic fibers used to fabricate a yellow blanket found in appellant's bedroom; 5) rayon and nylon fibers used to fabricate the carpet of a white 1970 Chevrolet station wagon to which appellant had access during part of the period over which the crimes in question occurred; 6) blue acrylic fibers used to fabricate a blue throw rug found in the porch or garage area of appellant's home; 7) polypropylene fibers used to fabricate a carpet located in a workroom in the back of appellant's home that is adjacent to appellant's bedroom; 8) yellow nylon, blue rayon, white polyester, and pigmented polypropylene fibers, for which no source from appellant's environment was identified, and which were recovered from vacuum sweepings made by the state of appellant's 1970 station wagon; 9) fibrous debris removed from a vacuum cleaner found in appellant's home; 10) white polypropylene fibers used to fabricate the trunk liner of a 1978 Plymouth Fury to which appellant had access during part of the period over which the crimes in question occurred; 11) white acrylic and secondary acetate fibers used to fabricate the trunk liner and red tri-lobal nylon fibers used to fabricate the interior carpet of a burgundy-colored 1979 Ford LTD to which appellant had access *757 during part of the period over which the crimes in question occurred; 12) blue secondary acetate fibers representing the composition of a bedspread taken from the porch or garage area of appellant's home; 13) brown woolen and rayon fibers which composed the lining of a leather jacket owned by appellant; 14) gray acrylic fibers used to

fabricate a gray glove which was found in the glove compartment of appellant's 1970 station wagon; 15) yellow nylon fibers which were used to fabricate a toilet seat cover taken from the Williams home and which were found in the fibrous debris vacuumed from the 1970 station wagon; and 16) yellow acrylic fibers used to fabricate a carpet which was found in the kitchen of appellant's home.

As to the carpet found in the appellant's bedroom (see #2 in the preceding paragraph), the state established that it was somewhat unique. A textile technologist named Henry Poston, who was the director of technical services for Wellman, Incorporated, a Boston, Massachusetts manufacturer of synthetic textile fibers, testified that he had begun working for Wellman in 1967, and that one of the first things he was asked to do was to assist in the development of a synthetic fiber known as the 181-b. According to Poston, this fiber had an unusual shape, trilobal with two long lobes and one short lobe, which was designed to avoid infringing upon a patented DuPont equilateral trilobal shape. The witness was shown state's exhibit #616, which was identified as a scanning electron microscope photograph of a fiber from the green carpet in appellant's bedroom, and he said it appeared to be a Wellman 181-b fiber. The state also called to the stand Gene Baggett, an employee of West Point Pepperell, a Dalton, Georgia carpet manufacturing company. Baggett was the company's purchasing manager for fibers, yarns, and backings, and he testified that his company had purchased the Wellman fibers in 1970 and 1971. He testified that the company had used the Wellman 181-b fiber to manufacture several lines of carpet, including lines known as Luxaire and Dreamer, both of which, he testified, had been colored with a dye formulation dubbed English Olive.

In the course of his testimony Baggett identified state's exhibit #623 as a sample of the Luxaire English Olive which had been obtained from his company's business records. He then was asked to examine state's exhibit #624, which was a sample of appellant's green bedroom carpet. Although he admitted that he was not a chemist, and was not qualified to perform microscopic analysis and identification of single fibers, he testified that, based upon his visual inspection of such aggregate physical characteristics as height of pile, weight of carpet, and type of backing, #624 appeared to be similar to #623.

Harold Deadman later testified that he had examined state's *758 exhibit #659, which was another piece of the green carpet from the bedroom of Wayne Williams, and state's exhibit #660, which was another piece of green carpet. He said that the FBI had obtained the latter exhibit from West Point Pepperell, which had identified it as a piece of Luxaire. Deadman testified that, based on his examination of the gross physical characteristics of the two exhibits, he could find no significant differences in their construction, and concluded that

"in all probability they were manufactured by the same company. They certainly could have come from the same source."

The state also adduced testimony concerning the cars to which appellant had access during the period over which the crimes occurred. Ralph Barnhart, appellant's uncle, testified that he loaned his 1970 white Chevrolet station wagon to appellant's father in October 1980, and that the Williams family had had exclusive possession of the car since that time. Detective Milton Jones of the Atlanta Bureau of Police Services testified that he had determined that the Williams family had been in possession of a 1979 burgundy-colored Ford LTD from May 15, 1979, when appellant's father purchased it new, to December 4, 1980, when it was repossessed. Jones testified that, when the car was repossessed, the company which repossessed it listed the mileage as 54,343. Jones also testified that he had determined that appellant bought a new 1978 blue Plymouth Fury on May 24, 1978, and owned it until it was repossessed on December 31, 1979, at which time the odometer registered 60,000 miles. Jones also testified that appellant's family rented a 1979 chamois-colored Fairmont station wagon on February 11, 1981, and returned it after it had been driven 104 miles.

We now turn to a recital of facts a jury would have been authorized to find from the evidence presented on the homicides of Jimmy Ray Payne and Nathaniel Cater, the two crimes with which appellant was charged.

Payne was 21 years of age, unemployed, and had no automobile or driver's license. A product of a broken home, he lived with his mother, sister and girl friend. The late morning of April 21, 1981, was the last time Payne was seen by any member of his household. It was then he told his mother he was on his way to the Omni.

The following day a witness saw Williams and Payne standing by a taxi cab which was stopped on Highway 78 approximately one mile from the Chattahoochee River. The witness saw Williams and Payne talking to the driver of the taxi cab, and he also saw a white station wagon parked on the opposite side of the street from the cab.

Payne's body was discovered clad only in red shorts in the Chattahoochee River on April 27, 1981. The medical examination and autopsy resulted in opinion evidence that the cause of death was *759 asphyxia by an undetermined method.

The state adduced testimony of seven fiber and hair associations between Wayne Williams and Jimmy Ray Payne. Larry Peterson testified 1) that two pale violet acetate fibers removed from Payne were consistent with violet acetate fibers present in the bedspread of

Williams, except that they were lighter in color; 2) that three green Wellman-type fibers removed from Payne's shorts were similar to and could have originated from appellant's bedroom carpet, again, except that they were lighter in color; 3) that a blue-green or blue-gray rayon fiber removed from Payne was consistent with the rayon fibers comprising the carpet of the 1970 station wagon; 4) that several light yellow rayon fibers and a light yellow acrylic fiber found on Payne were consistent with fibers composing the yellow blanket found in appellant's bedroom, except that they were lighter in color; and 5) that a blue acrylic fiber removed from Payne was consistent with the blue acrylic fibers which composed the blue throw rug found in appellant's bathroom. Harold Deadman testified 1) that a blue rayon fiber removed from Payne was consistent with blue rayon fibers, for which no source was known, found in various fibrous debris removed from the Williams home, and 2) that the approximately seven animal hairs removed from Payne could have originated from appellant's German Shepherd dog. There was evidence that the fibers found on Payne which were lighter in color than their supposed counterparts from the Williams environment were lighter because of their exposure to river water.

Nathaniel Cater was 28 years old, lived at the Falcon Hotel in downtown Atlanta, and did not own an automobile. Robert Henry, a friend of Cater, saw Cater holding hands with Wayne Williams outside the Rialto Theatre about 9:00 to 9:15 p. m. on May 21, 1981. About 3:00 a. m., May 22, 1981, a member of a police surveillance team stationed at the Jackson Parkway Bridge heard a loud splash in the Chattahoochee River and saw a circle of waves form on the water. An automobile was then observed starting up and crossing the bridge. When the car was stopped, it was found to be a white Chevrolet station wagon and Wayne Williams was the driver.

Cater's body was discovered in the Chattahoochee River on Sunday, May 24, 1981. It was located about 200 yards downstream from Interstate Highway 285 (the Cater body was found only a short distance from the location at which the Payne body was found). The medical examination and autopsy of the body revealed Cater weighed about 146 pounds and that his death was caused by asphyxia due to some kind of chokehold formed with a broad, soft surface such as a forearm.

Cater's body was nude; therefore, only his pubic and head hair *760 regions were capable of holding fiber or hair evidence. Even so, several fibers and hairs were recovered. Larry Peterson testified 1) that two pale violet acetate fibers removed from the head hair of Cater had the same characteristics as the violet acetate fibers present in Williams' bedspread, except that they were lighter in color; 2) that a green nylon fiber removed from Cater's head hair had similar characteristics and properties as the fibers which composed the carpet in

appellant's bedroom, except that it was lighter in color; 3) that a green polypropylene fiber taken from Cater's pubic hair had the same microscopic and optical characteristics as the fibers which composed the carpet in the workroom in the Williams home; 4) that a melted nylon fiber removed from Cater's head hair was consistent with nylon fibers found in the fibrous debris vacuumed from appellant's 1970 station wagon; 5) that a yellow rayon fiber removed from Cater's hair was consistent with the properties of the fibers present in the yellow blanket found in appellant's bedroom, except that it was lighter in color; and 6) that four animal hairs recovered from Cater were consistent with the characteristics of the hair of Williams' dog. There was evidence that the fibers found on Cater which were lighter in color than their supposed counterparts in the Williams environment were lighter because of their exposure to river water.

We turn now to the evidence relating to the ten independent homicides.

I). ALFRED EVANS:

The body of Alfred Evans was discovered on July 28, 1979 on Niskey Lake Road, about one mile from its intersection with Campbellton Road, a four lane highway. Evans, a fourteen-year-old, was 5'4-1/2" tall and weighed about 80 to 90 pounds at the time of his death. The body was clothed only in slacks, and Detective Lloyd of the Atlanta Bureau of Police Services testified that there was no evidence of a struggle at the scene, and that the body appeared to have been dumped there, as it was lying face down about eleven feet off the road on a steep decline. Dr. John Feegel, the Associate Medical Examiner for Fulton County, testified that the cause of death was probably asphyxia, with the form of asphyxia probably being strangulation. He found no injuries in the neck area.

Vera McDaniel, a neighbor of Evans, testified that Evans did not have access to a car and usually rode a bus for transportation, and that he did odd jobs around the housing project in which he lived. There was testimony that Evans was frequently out late.

The state adduced testimony of four fiber and hair associations between Williams and Evans. Agent Deadman testified 1) that two violet acetate fibers removed from Evans exhibited the same *761 microscopic and optical properties as violet acetate fibers removed from the bedspread of appellant; 2) that a fiber removed from Evans exhibited the same microscopic and optical properties as the Wellman fibers present in the carpet in appellant's bedroom and could have originated from that carpet; 3) that six polypropylene fibers found on Evans could have originated from the trunk liner of the Williams' 1978 Plymouth Fury; and 4) that animal hairs removed from Evans could have originated from appellant's dog. Larry Peterson's hair and fiber findings were consistent with Deadman's.

II). ERIC MIDDLEBROOKS:

On May 19, 1980, the fully clothed body of fourteen-year-old Eric Middlebrooks was discovered just off Flat Shoals Road, less than 1/2 mile from Interstate 20 and near both Moreland Avenue and Memorial Drive. He was approximately 4'10" tall and weighed about 88 pounds at the time of his death. Dr. Feegel performed the autopsy. He testified that there were two cuts or lacerations on the right side of the head that had caused significant external bleeding as well as internal hemorrhaging under the scalp. There was also a stab wound on the right upper arm, and a superficial stab wound in the left upper chest area. Dr. Feegel concluded that the death was homicidal and was probably caused by blows to the head by some kind of blunt instrument. On cross, after viewing a photograph which Dr. Feegel agreed showed a significant pool of blood around the victim's body, Dr. Feegel testified that he believed Middlebrooks died where he was found. However, he also testified that Middlebrooks could have been hit with a blunt instrument elsewhere, and then dumped where his body was found.

According to the testimony of the detective who investigated the Middlebrooks case, Middlebrooks lived just on the other side of Interstate 20 from where his body was found, a distance of approximately one mile. He also testified that Middlebrooks was last seen riding his bicycle around 10:00 p. m. the night before his body was found. At that time Middlebrooks was going to a store on Memorial Drive on an errand for one of his neighbors. The deceased's bicycle was found next to his body; his pants pockets had been turned inside out; and no money was found on the body.

Eric's older brother, Kerry, an officer with the Atlanta Police Department, testified that Eric lived with his foster father, Robert Miller, at the time of his death. According to Kerry, he and Eric began living with Robert and Lucille Miller in 1965, and had had a good family life since that time. Kerry also testified that Eric used his bicycle for transportation, ran errands and did odd jobs for people in the neighborhood, and kept fairly late hours.

*762 Agent Deadman testified as to five fiber and hair associations between Middlebrooks and Williams. He testified 1) that four violet acetate fibers removed from Middlebrooks were consistent with having originated from appellant's bedspread; 2) that thirty-two red nylon fibers that were found in a clump on one of his shoes could have originated from the interior carpet of the 1979 Ford LTD; 3) that two white acrylic and two secondary acetate fibers found on Middlebrooks could have originated from the trunk liner of the 1979 Ford; 4) that one yellow nylon fiber found on Middlebrooks could have originated from either the toilet cover in the Williams home or from the same source (which he could not identify) that produced the loose yellow nylon fibers that were found in the debris vacuumed from

the 1970 Chevrolet station wagon; and 5) that one animal hair removed from Middlebrooks could have originated from the Williams dog. Larry Peterson's testimony concerning the hair and fiber comparisons he had conducted was substantially to the same effect as Deadman's.

III). CHARLES STEPHENS:

On October 10, 1980, the partially clothed body of twelve-year-old Charles Stephens was found near the rear entrance to a mobile home park. The body was clothed only in blue jeans and one tennis shoe. The detective who reported to the crime scene testified that it appeared the body had been laid out, that there were no signs of a struggle, and that no motive could be established for the killing. The victim's grandfather testified that Stephens lived with his mother in a public housing project and often ran errands for people in the neighborhood to make money. There was also testimony that the victim frequently hung around in the streets.

Dr. Feegel examined the body. He testified that Stephens was about 4'9" inches tall and weighed about 108 pounds. Because petechial hemorrhages, which are signs of asphyxia, were located on several parts of the body, he thought the cause of death was asphyxia. And, because of the lack of injuries around the neck, he concluded that the form of asphyxia was suffocation.

The state adduced testimony of eight fiber and hair associations between Stephens and Williams. Agent Deadman testified 1) that thirty-five violet acetate and a number of green cotton fibers found on Stephens could have originated from the bedspread found on appellant's bed; 2) that three yellow nylon fibers removed from Stephens could have originated from the carpet found in appellant's bedroom; 3) that two polypropylene fibers found on Stephens could have originated from the workroom in the back of the Williams home; 4) that about thirty undyed synthetic and about 20 secondary acetate *763 fibers recovered from Stephens were consistent with having originated from the trunk liner of the 1979 Ford LTD; 5) that nine blue rayon fibers found on Stephens were similar to blue rayon fibers, the source of which was unknown, found in debris vacuumed from the 1970 station wagon, debris removed from the sweeper found in the Williams home, and debris removed from the bedspread found in appellant's bedroom; 6) that one yellow nylon fiber taken from Stephens could have originated from the toilet cover found in the Williams home, or from the same source, which was unknown, that produced the yellow nylon fibers found on some of appellant's clothing in the debris removed from the 1970 station wagon; 7) that five coarse white polyester fibers removed from Stephens could have originated from the same source, which was unknown, that produced the white polyester fibers removed from a

white rug found in appellant's 1970 station wagon; and 8) that the approximately seventeen animal hairs found on Stephens could have originated from appellant's dog. Again, Larry Peterson substantially corroborated Deadman's testimony.

IV). TERRY PUE:

On January 23, 1981, fifteen-year-old Terry Pue's fully clothed body was discovered. It was lying approximately one half mile from Interstate 20 about two feet off the pavement of a Rockdale County road. James Dawson, a medical examiner for the Georgia Crime Laboratory, performed the autopsy on Pue's body. He testified that the body was 5'5-1/2" tall, and weighed approximately 100 lbs. He also testified that he found ligature marks around the neck, as well as some hemorrhaging underneath the skin in the neck area. He concluded that the cause of death was asphyxia, definitely by ligature, and possibly by manual, strangulation.

Pamela Terrell, Terry Pue's sister, testified that Terry worked various odd jobs, and hung out at various places around the city, including East Point, East Lake, and the Omni. She last saw him three days before his body was found.

A friend of Pue's Charmaine Kendrick, testified that she saw Pue with Wayne Williams about a week before she learned of the discovery of Pue's body. Kendrick worked at a Church's Fried Chicken, and, according to her testimony, Pue would occasionally come by the store to sell odds and ends. She testified that, late one afternoon, Pue came by the store and tried to sell her a water gun, and that, when she asked him where he would get it, he responded that he worked for a man. He then left, she testified, walked toward a green station wagon, and talked to a man sitting in that car, a man, Kendrick said, whom Pue had identified as the man for whom he *764 worked. At trial she identified this man as Wayne Williams.

Several witnesses also placed Williams at the crime scene and at Pue's funeral. Two police officers who were at the crime scene testified that Williams came to the crime scene, presented some identification, and asked if he could take pictures. Two witnesses also testified that they saw Williams at the funeral of Terry Pue.

The state adduced testimony of five fiber and hair associations between Pue and Williams. Agent Deadman testified 1) that over one hundred violet acetate and a number of green cotton fibers found on Pue were all consistent with having originated from the bedspread found in appellant's bedroom; 2) that three yellow nylon fibers found on Pue could have originated from the carpet located in appellant's bedroom; 3) that two pale green

polypropylene fibers removed from Pue could have originated from the carpet located in the workroom in the back of the Williams home; 4) that one coarse white polyester fiber recovered from Pue had the same properties as white polyester fibers, the source of which was unknown, vacuumed from the rug and interior of appellant's 1970 station wagon; and 5) that the approximately seventeen animal hairs found on Pue could have originated from the Williams dog.

V). LUBIE GETER:

The body of fourteen-year-old Lubie Geter, clad only in underpants, was discovered on February 5, 1981, in a wooded area about seventy feet off Vandiver Road.

Dr. Robert Stivers, Chief Medical Examiner for Fulton County, performed the autopsy on Geter's body. He testified that the body was 5'7" tall and weighed approximately 103 lbs. Because of petechial hemorrhages found on the heart, lungs, and brain, he concluded the death was definitely asphyxial. And, because of hemorrhages found under the skin in the neck area, he testified that the form of asphyxia was strangulation, with the method of strangulation probably being a chokehold.

Geter lived with his mother and father, and, according to his uncle's testimony, he did odd jobs, such as yard work and washing cars, to make money. Geter was reported missing on January 3, 1981, and his mother testified that the morning of his disappearance, January 2, 1981, which was the last time she saw him, he was leaving home to sell car deodorizers. His mother testified that for transportation he rode his minibike and the bus system.

Eric Conway, who had known Geter for six or seven years, testified that on the morning of January 2, 1981, Franklin Jordan, Geter's half brother, drove him and Geter to the Stewart-Lakewood area, where they were going to sell car deodorizers. He testified that Lubie had a box of about twenty-three deodorizers, but that because *765 he did not have any, he did not stay at the mall but instead went home with Jordan. He testified that they let Lubie out in front of a Big Star grocery store, and that Lubie said he would call Jordan to come get him later in the day.

Franklin Jordan testified, and confirmed that he had driven Lubie and Eric Conway to a shopping center in the Stewart-Lakewood area. He testified that he dropped Lubie in front of a Big Star and that Eric left with him. Jordan testified that that was the last time he saw Lubie and that Lubie never called him to come pick him up.

Ruth Warren, a resident of Rockdale County, testified that on January 2, 1981 she drove to Atlanta to take her mother shopping. She testified that she and her mother first went to a

Kroger grocery store on Stewart Avenue, about a block from the Big Star on Stewart. She testified that a boy, whom she identified as Lubie Geter, approached her in front of the Kroger at about 1:30 p. m. and asked her if she wanted to buy an air freshener. From the Kroger, she testified, she and her mother went to the Royal Mattress Company, then to her mother's home to put away the groceries and then back to the mattress company, arriving there around 3:00 p. m. She testified that the mattress company was located next to the Stewart-Lakewood Shopping Center, on the side opposite the Kroger store. According to Warren, her mother picked out a mattress, and, when she went outside to open her van door so the mattress could be loaded, she saw, about twenty feet from her, a black man and boy talking. She testified that she heard the child tell the man that he would like to go with him, but that he needed to sell what he had in a box. Warren said that she then saw them walk away, and that she thought they were walking over towards a Zayre's store or towards a green car that was parked in the parking lot. She did not see them get in any vehicle. At trial, she identified the boy as Lubie Geter and the man as Wayne Williams. On cross, she testified that the car they were walking towards looked similar to her Ford station wagon. After Geter's disappearance, Ms. Warren talked with the police about the man she had seen with Geter, and at that time she told the police that the man was not wearing glasses.

A fifteen-year-old juvenile also placed Lubie Geter with Wayne Williams on January 2, 1981. He testified that on that day he was working at a carpet store in the Stewart-Lakewood area. The witness stated that, when he was working on a loading dock, he saw Lubie Geter, whom he did not know very well, get into a car with a man he identified at trial as Wayne Williams. He testified that the car Geter and Williams drove away in was white with a black top. The witness also testified that in August of 1980 he was in the Stewart-Lakewood *766 area, and that Williams asked him if he wanted a job. He testified that he answered that he did, and that he got in the car with Williams, and that Williams then drove around town and fondled him, and finally stopped the car saying that he needed to get something out of the trunk. The juvenile stated that after Williams got out of the car, he jumped out and ran to a nearby apartment complex. He also testified that he saw Williams at the funeral of Lubie Geter.

There was testimony as to five fiber and hair associations between Geter and Williams. Agent Deadman testified 1) that several violet acetate fibers found on Geter were consistent with having originated from the bedspread found in appellant's bedroom; 2) that five yellow nylon carpet fibers removed from Geter had the same characteristics as the fibers present in the carpet located in appellant's bedroom; 3) that one yellow acrylic fiber discovered on Geter could have originated from a carpet found in the kitchen of the Williams home; 4) that a green rayon fiber found on Geter could have originated from the

carpet of appellant's 1970 station wagon; and 5) that ten animal hairs removed from Geter could have come from appellant's dog. In his testimony, Larry Peterson substantially corroborated Deadman's findings.

VI). PATRICK BALTAZAR:

On February 13, 1981, the fully clothed body of eleven-year-old Patrick Baltazar was discovered in a wooded area just off a road running through Corporate Square, an office complex located about three blocks from Interstate 85. Baltazar was about 5'5" tall and weighed about 130 pounds at the time of his death. Dr. Joseph Burton, the DeKalb County Medical Examiner, testified that, because of hemorrhages he found, and because of ligature marks on the neck, he thought the cause of death was asphyxia due to ligature strangulation.

A DeKalb County Police Department investigator, B. W. Humble, testified that Baltazar lived near the Omni, in an apartment on Foundry Street, but that he frequently roamed around town, and, in particular, often visited his father where he worked on Courtland Street, and that for transportation Baltazar walked or used the bus system. Additionally, in response to the question whether Baltazar came from a broken family, Humble testified that Baltazar's mother lived in Louisiana, and that his father, although listing his address as Foundry Street, actually lived with his girl friend in an apartment several miles from Foundry Street and only infrequently returned to that address. A friend of Baltazar's also testified that Baltazar kept late hours and worked odd jobs to make money.

With regard to hair and fiber associations between Williams and Baltazar, testimony of eleven such associations was adduced. Agent *767 Deadman testified 1) that violet acetate and green cotton fibers removed from Baltazar were consistent with having originated from appellant's bedspread; 2) that seven yellow nylon Wellman-type fibers that were removed from Baltazar exhibited the same characteristics and properties as fibers present in the carpet located in appellant's bedroom and could have originated from that carpet; 3) that four yellow rayon fibers removed from Baltazar's jacket could have come from the yellow blanket found in appellant's bedroom; 4) that four rayon fibers found on Baltazar, which, because of deterioration, ranged from green to yellow in color, could have originated from the carpet of appellant's 1970 station wagon; 5) that two woolen fibers and one rayon fiber found on Baltazar exhibited the same characteristics as woolen and rayon fibers taken from the cloth waistband of appellant's leather jacket; 6) that thirteen gray acrylic fibers removed from the t-shirt, jacket, and shirt of Baltazar could have originated from the gray glove that was found in the glove compartment of appellant's 1970 station wagon; 7) that a light yellow nylon fiber, 8) a coarse white polyester fiber, and 9) a pigmented

polypropylene fiber had the same properties as fibers present in the debris vacuumed from the 1970 station wagon, and could have originated from the same sources, which were unknown, that produced the fibers discovered in the debris; 10) that the approximately twenty animal hairs found on the clothing of Baltazar could have come from the Williams dog; and 11) that two scalp hairs removed from Baltazar were inconsistent with Baltazar's own scalp hair, but were consistent with scalp hairs taken from Williams, and could have originated from appellant.

Both Gaudette and Peterson reached substantially the same conclusions on their own hair and fiber comparisons as Deadman in Baltazar's case.

VII). LARRY ROGERS:

On April 9, 1981, the body of eighteen-year-old Larry Rogers was discovered in an abandoned apartment, just off Simpson Street and a little less than a mile from the Bankhead Highway. Rogers was clothed in shorts and tennis shoes. Dr. Robert Stivers testified that the body was 5'2" long and weighed 130 pounds. Because of hemorrhaging around the neck, heart, and lungs, and because the thin bones at the back of the voice box were broken, Dr. Stivers thought the cause of death was asphyxia by strangulation, possibly by a chokehold.

A neighbor of Rogers, Marion Butler, testified that Rogers lived with foster parents and worked odd jobs around the neighborhood, such as raking leaves and cleaning gutters. Rogers played ball, Butler said, with young children at various parks and schools in the area, *768 and, according to Butler, Rogers, although old enough to drive, did not have a car and rode his bicycle or a bus for transportation. There was testimony that Rogers was slightly retarded.

Tilbert Baynham, who said he had known Rogers about a year and a half before his disappearance, testified that he saw Rogers three times one day about three days before Rogers disappeared. He said that he first saw Rogers that day at about 9:30 a. m. at a fast food place on Simpson Street. According to Baynham's testimony, Rogers approached him and asked him if he could supply Rogers with a joint so that Rogers and a friend could get high. Baynham testified that Rogers pointed out his friend, and that his friend was sitting in a fast-looking car, the color of which Baynham could not recall. According to Baynham, he then got Rogers a joint, and Rogers got in the car with his friend and left. At trial Baynham identified the deceased's friend as Wayne Williams.

Baynham testified that he saw Rogers and Williams again that day between 2:00 p. m. and 3:00 p. m. At that time, Baynham testified, Williams and Rogers were in a car which was stopped at a traffic light at the corner of West Lake and Simpson streets. Baynham testified that Williams was driving the car, and that both Williams and Rogers waved at him. Baynham also testified that between 6:00 p. m. and 7:30 p. m. that day he saw Williams and Rogers driving in the same area.

Nellie Trammell, a friend of Rogers, who had known him since he was five or six years old, also placed Wayne Williams with Rogers about the time of the victim's disappearance. She testified that she last saw Rogers on Monday, March 20, 1981, and that that Thursday she heard on the news that he had been reported missing. On that Monday, Trammell testified, she was driving home about noon, when a green station wagon, with Williams and Rogers in it, cut in front of and then came to a stop beside her. She said that Williams was driving, and that she spoke to Rogers, who was slumped over, but that he did not say anything. Trammell then drove home. She testified that she later saw Williams at the victim's funeral.

The state adduced testimony of seven fiber and hair associations between Williams and Rogers. Agent Deadman testified 1) that thirteen violet acetate fibers removed from Rogers were consistent with the violet acetate fibers taken from the bedspread of appellant; 2) that three yellow-green nylon fibers removed from Rogers were similar to the Wellman fibers found in appellant's bedroom carpet; 3) that eight yellow rayon fibers discovered on Rogers could have originated from the yellow blanket found in appellant's bedroom; 4) that one yellow-brown to green fiber taken from Rogers could have come from the carpet of the 1970 station wagon; 5) that two secondary acetate fibers removed from the deceased's shorts could have *769 originated from the bedspread that was found in the Williams garage; 6) and that a light yellow nylon fiber removed from the head hair of Rogers exhibited the same characteristics as yellow nylon fibers removed from the toilet cover found in appellant's home, from the sweepings made of the 1970 station wagon, and from several items of clothing of appellant. Additionally, Larry Peterson testified that animal hairs found on Rogers could have originated from appellant's dog.

VIII.) JOHN PORTER:

On April 12, 1981, the fully clothed body of twenty-eight-year-old John Porter was found near Capitol Avenue in down-town Atlanta, approximately one mile from Interstate 20 and three miles from Interstate 85. Dr. Stivers testified that Porter's body was 5'10" long and weighed 123 lbs. Dr. Stivers testified that Porter died from six stab wounds to the chest and abdomen, all of which were approximately 2 to 2-1/2 inches deep. In response to a question on direct, which inquired whether Porter would have bled when stabbed, Dr. Stivers

responded that, because of the places where Porter was stabbed, all of the bleeding Porter did was internal. Stivers also testified that there were no marks or hemorrhaging around the neck area.

It was the opinion of one of the investigators that no foul play occurred at the scene at which the body was discovered, Porter having been dumped there after he was killed. In this regard, the officer testified that, although the victim had been stabbed, there was no blood at the scene. This investigator also testified that he had determined that Porter lived alone in an abandoned apartment on Capitol Avenue, and that he did not live with his parents because of problems he had with his mother. He had recently been released from the Georgia Regional Hospital. He was unemployed and had no vehicle.

Agent Deadman testified as to seven fiber and hair associations between Porter and Williams. He testified 1) that violet acetate and green cotton fibers found on Porter could have originated from the bedspread of appellant; 2) that one yellow-green nylon fiber removed from the sheet used to carry Porter exhibited the same characteristics as the Wellman fibers making up appellant's bedroom carpet and could have originated from that carpet; 3) that three yellow rayon fibers removed from Porter matched the yellow rayon fibers removed from the blanket found in appellant's bedroom; 4) that several green rayon fibers removed from Porter could have originated from the carpet of the 1970 station wagon; 5) that two secondary acetate fibers removed from Porter could have originated from the bedspread found in the carport of the Williams home; 6) that a blue rayon fiber *770 found on Porter could have come from the same source, which was unknown, that produced the blue rayon fibers found in the debris removed from the 1970 station wagon and in the debris removed from the vacuum cleaner found in appellant's home; and 7) that the approximately seven animal hairs removed from Porter were consistent with having originated from the Williams dog.

The state also attempted to link Porter with Williams through a blood stain found on the rear car seat of appellant's 1970 station wagon. Forensic serologists from the Georgia Crime Laboratory examined a blood sample from Porter, and determined that his blood type was International Blood Group B and that his blood enzyme type was PGM-1, a combination, a serologist testified, which exists in approximately seven percent of the population. Another blood stain found on the car seat was determined to be blood from International Blood Group B, with an enzyme type PGM-1, and a serologist testified that this blood stain was not more than eight weeks old. Moreover, a serologist testified that Williams could not have left this blood stain as his blood type was International Blood Group O.

IX). JOSEPH BELL:

On April 19, 1981, the body of fifteen-year-old Joseph Bell was discovered in the South River near the DeKalb and Rockdale County lines. The body was clothed only in a pair of underwear. Dr. Burton performed the autopsy on Bell's body, and he testified that the body was in a fairly advanced state of decomposition, which was evidenced by partial skeletization of the skull, hands, feet, and chest, by distention of the abdomen, and by several generations of flies and maggots present on the body. Based on the state of decomposition, Burton testified that he thought Bell would have had to have been dead since the early part of March 1981. He testified that, although the condition of the body made measuring and weighing it difficult, he determined that the body was about 5'4" in length, and that Bell would have weighed about 120 pounds when alive.

As to bodily injuries, Burton testified that he found evidence of antemortem hemorrhaging behind the Adam's apple and in the high part of the neck, just in front of the spine. He testified that there was no water in the sinuses or airways unlike the usual condition of a drowning victim. Because of these findings, and because of the lack of evidence of any other injuries, Burton thought the cause of death was asphyxia due to some kind of manipulation of the neck area.

John Laster, a friend of Bell, testified that he had met Wayne Williams at his grandmother's house, that he had auditioned before him, and that Joseph Bell had also auditioned before Williams. Laster also testified that Joseph lived with his mother, brother, and two sisters, and that Joseph played basketball at various gyms *771 around Atlanta.

John Laster's older brother, Lugene, testified that in November of 1980 he saw Williams talking to John at his grandmother's house. Lugene also testified that, around the first of March 1981, he was playing basketball with Joseph Bell at an elementary school, and that, when Joseph left and began walking down the street that ran beside the court, he saw Joseph stop and talk to, and then get in the car with, a man driving an old white or sky blue station wagon. At trial Lugene identified this man as Wayne Williams. According to Lugene's testimony, Joseph Bell's brother came to his home that night asking if anyone had seen Joseph.

Another witness connected Joseph Bell and Wayne Williams. Kent Hindsman, who was twenty-four years old at the time of trial, testified that he met Williams as a result of picking up a flyer Williams had left at a record store, which flyer requested that aspiring musicians get in touch with Wayne Williams. Because Hindsman was interested in music production, he called Williams, and arranged to go to a studio with him. Hindsman testified that this visit occurred on December 8, 1980, and that, when he went back to the studio about one week later, he met Joseph Bell. After that session, according to

Hindsman, Williams drove him and Bell home in a white station wagon. Hindsman also testified that, when at one point he broached the subject of missing children with Williams, Williams told him they "ought to keep their damn asses at home," and that, on another occasion, Williams passed him a note at the studio which read, "I could be a President, I could be a mayor, or I could even be a killer."

The state adduced testimony showing two fiber associations between Bell and Williams. Agent Deadman testified 1) that five blue rayon fibers found on Bell were similar to rayon fibers recovered from debris collected from the 1970 station wagon and from debris collected from appellant's bedspread; and 2) that two pale violet acetate fibers found on Bell were consistent with the fibers present in the bedspread of Williams, with the exception that they were considerably lighter in color. Deadman attributed this paleness of the two fibers to exposure to river water, basing his opinion on an experiment wherein the state had taken fibers from appellant's bedspread and placed them in water from both the Chattahoochee and South rivers; the water had bleached the fibers, causing their color to fade.

X). WILLIAM BARRETT:

On May 12, 1981, the fully clothed body of sixteen-year-old William Barrett was discovered just off Winthrop Road, within one hundred feet of Interstate 20. Dr. Joseph Burton, who performed the autopsy on Barrett's body, testified that Barrett could have been *772 dead as little as 3 or 4 hours or as long as 12 to 18 hours, and that the corpse was about 5'5" in length and weighed about 130 lbs. Because Dr. Burton found ligature marks around the neck and hemorrhages within the neck and on the eyes, heart, and lungs, he testified that the cause of death was asphyxiation by ligature strangulation. Dr. Burton also testified that Barrett had two stab wounds in the right lower part of the abdomen, which had not caused any internal or external bleeding, and which had occurred postmortem and were definitely not the cause of death. He further testified that there was a one-inch laceration on Barrett's scalp, which had caused some bleeding beneath the scalp, and which he felt had occurred when the body was tossed to the ground. Additionally, Dr. Burton was of the opinion that Barrett had not been killed where his body was found, but that he had been killed elsewhere and then dumped along the side of Winthrop Road.

James Barrett, William's cousin, testified that Barrett lived only with his mother, worked odd jobs, and rode his bicycle or a bus for transportation. James also connected appellant with William Barrett. James testified that one night, sometime in late 1980 or early 1981, William and two friends came by his house, and that William asked to borrow ten dollars. James testified that he gave William the money, and that William and his two friends left.

At trial James identified one of William's friends as Wayne Williams. The testimony of this witness was substantiated by his mother, who testified that, late in 1980, William Barrett brought Wayne Williams, along with a few other boys, into her home.

The state adduced testimony of seven fiber and hair associations between Barrett and the appellant. Agent Deadman testified 1) that the many violet acetate and green cotton fibers removed from Barrett could have originated from the appellant's bedspread; 2) that five yellow-green nylon fibers recovered from Barrett could have originated from the appellant's bedroom carpet; 3) that seven yellow rayon fibers removed from Barrett could have originated from the blanket found under William's bed; 4) that a blue rayon fiber recovered from Barrett had the same characteristics as blue rayon fibers recovered from the debris removed from the station wagon, from the vacuum cleaner found in appellant's home, and from the appellant's bedspread; 5) that approximately thirty gray acrylic fibers recovered from Barrett could have originated from the glove recovered from the glove compartment of the appellant's 1970 station wagon; 6) that three fibers removed from Barrett could have originated from the carpet of the 1970 station wagon; and 7) that the approximately thirteen animal hairs recovered from Barrett could have come from the appellant's dog.

*773 As with Porter, the state attempted to link William Barrett with Wayne Williams through blood stains found on the rear car seat of appellant's 1970 station wagon. Two blood stains found on this seat were determined to have originated from a person having a blood type of International Blood Group A, and a blood enzyme type of PGM-1, a combination which exists in only twenty-four percent of the population, and a combination which two state forensic serologists testified was found in a blood sample taken from William Barrett. Because, as has already been noted, appellant's blood type is International Blood Group O, he could not have left these two stains.

To better examine the admissibility of the above evidence, we have prepared a comparative profile of each of the homicides, charged and uncharged. These profiles illustrate the similarities of the victims and their deaths, the logical connection of all of the homicides and the evidence that Williams was the perpetrator of each.

VICTIM: Jimmy Ray Payne. PROFILE OF VICTIM: Age: 21 Sex: Male Race: Black Height: 5'7" Weight: 138 lbs. Home Situation: Lower income. Victim from broken home, living with mother, sister, and girl friend. Recently released from detention center. Habits: Unemployed. No car or driver's license. When last seen, victim was on his way to the Omni. CIRCUMSTANCES OF DEATH: Date body found: 4/27/81 Place body found: Chattahoochee River, 1/4 mile downstream from I-285 overpass. Condition of body: Clad

only in shorts. Cause of death: Asphyxiation by undetermined means. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with Williams' environment: 6 Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: Payne *774 disappeared 4/21/81. On 4/22/81 Payne seen with Williams on Highway 78, approximately 1 mile from Chattahoochee River and near a parked white station wagon. Sighting of Williams at crime scene or victim's funeral: No. VICTIM: Nathaniel Cater. PROFILE OF VICTIM: Age: 28 Sex: Male Race: Black Height: 5'10" Weight: 146 lbs. Home Situation: Lived in downtown Atlanta hotel on Luckie St. Habits: Worked out of labor pool occasionally. Drank a lot. Had no car. Frequented downtown lounges, also Rialto Theatre. CIRCUMSTANCES OF DEATH: Date body found: 5/24/81 Place body found: Chattahoochee River, 200 yds. downstream from I-285 overpass. Condition of body: Nude. Cause of death: Asphyxiation probably by chokehold. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 5. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: Two witnesses sighted the victim in the company of Williams. One saw him during the week before Cater disappeared. There was a station wagon containing a German Shepherd dog parked nearby. The other sighting, by the last person to see Cater alive, was of Williams coming out of the Rialto Theatre with the victim. The victim and appellant were holding hands. Sighting of Williams at crime scene or victim's funeral: Members of a surveillance team stationed on *775 Jackson Parkway Bridge over the Chattahoochee heard a splash around 3:00 a. m. 5/22/81 and saw a circle of waves form in the river. A white Chevrolet station wagon driven by Wayne Williams was observed to start up and cross the bridge and was subsequently stopped by police. VICTIM: Alfred Evans. PROFILE OF VICTIM: Age: 14 Sex: Male Race: Black Height: 5'4-1/2" Weight: 86 lbs. Home Situation: Lived in East Lake Meadows Housing Project. Habits: Had no car. Did odd jobs. Was frequently out late. CIRCUMSTANCES OF DEATH: Date body found: 7/28/79 Place body found: 11 ft. off Niskey Lake Road, north of Campbellton Rd. in southwest Atlanta. Body dumped down embankment. No sign of struggle at scene. Condition of body: Clad in slacks. Cause of death: Asphyxiation, probably strangulation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 3. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: No. Sighting of Williams at crime scene or victim's funeral: No. VICTIM: Eric Middlebrooks. PROFILE OF VICTIM: Age: 14 Sex: Male Race: Black *776 Height: 4'10" Weight: 88 lbs. Home Situation: Raised in foster home. Had lived with present foster father since 1965. Older brother officer with Atlanta Police Dept. Habits: Kept fairly late hours. Ran errands for neighbors. Transportation by bicycle. CIRCUMSTANCES OF

DEATH: Date body found: 5/19/80 Place body found: Off Flat Shoals Rd., less than 1/2 mile from I-20. Condition of body: Fully clothed. Cause of death: Blows to the head with blunt instrument. Two stab wounds. Body could have been dumped at site. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 4. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of William in company of victim: No. Sighting of Williams at crime scene or victim's funeral: No. VICTIM: Charles Stephens. PROFILE OF VICTIM: Age: 12 Sex: Male Race: Black Height: 4'9" Weight: 108 lbs. Home Situation: Came from broken home. Lived in public housing project with mother. Habits: Ran errands in neighborhood to make money. Hung around in the streets frequently. Had no car. CIRCUMSTANCES OF DEATH: Date body found: 10/10/80, day after he disappeared. Place body found: 5 miles from home, off Normandary Drive in East Point, Ga. *777 Condition of body: Laid out next to road, T-shirt, belt and socks were missing. No sign of struggle at scene. Cause of death: Asphyxiation: probable suffocation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 7. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: No. Sighting of Williams at crime scene or victim's funeral: No. VICTIM: Terry Pue. PROFILE OF VICTIM: Age: 15 Sex: Male Race: Black Height: 5'5-1/2" Weight: 100 lbs. Home Situation: Pue lived with his mother at a public housing project. Had no car. Had attended "challenge school," a home for juveniles in trouble. Habits: Hung out at the Omni and in West End game rooms. CIRCUMSTANCES OF DEATH: Date of body found: 1/23/81 Place body found: Near I-20 on Sigman Rd. in Rockdale County. Condition of body: Fully clothed. Body appeared "laid out." Cause of death: Asphyxiation, possibly by manual strangulation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 4. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: A witness saw Pue with Williams about a week before the witness learned that his body had been discovered. The *778 witness testified that Williams was in a green station wagon. Sighting of Williams at crime scene or victim's funeral: Two officers saw Williams at the scene where Pue's body was discovered. Williams had camera equipment and offered to shoot crime scene photos. Two witnesses saw appellant at victim's funeral. Williams was in a white station wagon. VICTIM: Lubie Geter. PROFILE OF VICTIM: Age: 14 Sex: Male Race: Black Height: 5'7" Weight: 103 lbs. Home Situation: Lived with parents. Habits: Did odd jobs to make money. Frequented the Omni. Last seen at a mall in the Stewart-Lakewood area in Atlanta. CIRCUMSTANCES OF DEATH: Date body found: 2/5/81 Place body found: Wooded area 70 ft. off Vandiver Road, which runs off Campbellton Road. Condition of body: Lying on its back with most clothing missing. Some of the clothing was found nearby.

Cause of death: Asphyxiation, probably strangled with a chokehold. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 4. Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: Two witnesses saw Williams with Geter on the day he disappeared. One originally said Geter was with a man not wearing glasses. The other testified Geter got into a car with man he identified as Williams. The car was identified as white with black top. This witness was a fifteen-year-old who testified that he had been approached by Williams in the same area the previous *779 August and offered a job. He got into the car with Williams, who fondled him as they drove around. This witness escaped when Williams stopped and got out saying he needed something from the trunk. Sighting of Williams at crime scene or victim's funeral: The juvenile witness testified that Williams was at the victim's funeral. VICTIM: Patrick Baltazar. PROFILE OF VICTIM: Age: 11 Sex: Male Race: Black Height: 5'4-1/2" Weight: 130 lbs. Home Situation: From broken home. Lived part time with father in area near Omni. Habits: Kept late hours. Did odd jobs to make money. Hung out at the Omni. Frequently on the streets in the area between Foundry Street and the Omni. Had no vehicle. CIRCUMSTANCES OF DEATH: Date body found: 2/13/81 Place body found: Corporate Square, an office complex located 3 blocks from I-85. Condition of body: Fully clothed but clothing unbuttoned. Appeared to have been dumped at the scene and rolled down an embankment. Cause of death: Probably asphyxiation due to ligature strangulation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 9 Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: No. Sighting of Williams at crime scene or victim's funeral: No. Additional associations: Two scalp hairs inconsistent with victim's hair but consistent with appellant's hair. *780 VICTIM: Larry Rogers. PROFILE OF VICTIM: Age: 18 Sex: Male Race: Black Height: 5'2" Weight: 130 lbs. Home Situation: Lived with foster parents. Habits: Used bike for transportation. Did odd jobs. Played ball with young children at various parks. Rogers was slightly retarded. CIRCUMSTANCES OF DEATH: Date body found: 4/9/81 Place body found: Abandoned apartment off Simpson St. less than 1 mile from Bankhead Hwy. Condition of body: Clad only in shorts and tennis shoes. Cause of death: Asphyxiation due to strangulation, possibly by chokehold. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 6 Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim. Two friends of the victim placed him with Williams before his disappearance. One saw him three times in one day about three days before he disappeared. Another friend saw him with Williams about the time of his disappearance in a green station wagon. She actually spoke to the victim who was slumped over and did not reply. Sighting of Williams

at crime scene or victim's funeral: The witness who saw Rogers with Williams on the date he disappeared also saw Williams at the victim's funeral. VICTIM: John Porter. PROFILE OF VICTIM: Age: 28 Sex: Male *781 Race: Black Height: 5'10" Weight: 123 lbs. Home Situation: Lived alone in abandoned apartment. Habits: Recently released from Ga. Regional Hospital. Unemployed. Had no vehicle. CIRCUMSTANCES OF DEATH: Date body found: 4/12/81 Place body found: Near Capitol Ave. approximately 1 mile from I-20, 3 miles from I-85. Condition of body: Fully clothed. No evidence of foul play at the scene. Body may have been dumped. Cause of death: Six stab wounds to chest and abdomen. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 6 Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: No. Sighting of Williams at crime scene or victim's funeral: No. Additional associations: Blood stain in appellant's white Chevrolet station wagon was found to match Porter's blood which was type B with enzyme PGM-1. This combination is found in 7% of the population. Williams has type O blood. VICTIM: Joseph Bell. PROFILE OF VICTIM: Age: 15 Sex: Male Race: Black Height: 5'4" Weight: approximately 120 lbs. Home Situation: Victim lived with mother and 8 other family members on Lawton St. in West End. Habits: Attended school infrequently. Played basketball at area schools in Mechanicsville and Adamsville. *782 CIRCUMSTANCES OF DEATH: Date body found: 4/19/81 Place body found: South River near the DeKalb and Rockdale county line. Condition of body: Clad only in undershorts. Body partly decomposed. Time of death set in early March 1981. Cause of death: No evidence of drowning. Cause of death set as asphyxiation due to some type of neck manipulation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 2 Presence of animal hairs consistent with hairs from appellant's dog: No. Sighting of Williams in company of victim: A friend testified that he had been with Bell when Bell auditioned before Williams. Another witness saw Bell get into a white or skyblue station wagon with appellant the last time Bell was seen alive at the school basketball court. A third witness had gotten a ride with appellant when Bell was in the car. Sighting of Williams at crime scene or victim's funeral: No. VICTIM: William Barrett. PROFILE OF VICTIM: Age: 16 Sex: Male Race: Black Height: 5'5" Weight: 130 lbs. Home Situation: Barrett lived with his mother and was on probation to the Dept. of Human Resources Youth Division. He was last seen by his court services worker prior to his appointment on 5/11/81. Habits: Worked odd jobs. Had no car. CIRCUMSTANCES OF DEATH: Date body found: 5/12/81 Place body found: Off Winthrop Rd. just off I-20. Condition of body: Body appeared to have been dumped on the road. Five knife *783 pricks in the body but only two holes in victim's shirt. Clothing was unbuttoned and pants were loose. In addition to knife pricks in the abdomen, there were two horizontal post-mortem

stab wounds. Cause of death: Asphyxia due to strangulation. EVIDENCE THAT WILLIAMS MURDERED VICTIM: Fiber association consistent with appellant's environment: 6 Presence of animal hairs consistent with hairs from appellant's dog: Yes. Sighting of Williams in company of victim: Barrett's cousin and aunt both testified to Barrett's bringing Williams to their home. Sighting of Williams at crime scene or victim's funeral: No. Additional associations: Two blood stains in Williams' 1970 station wagon were consistent with Barrett's blood type A with a blood enzyme type PGM-1, a combination existing in only 24% of the population.

On appeal Williams contends that the evidence of the 10 other offenses was inadmissible because each was a separate crime, unrelated to any other and unrelated to the murders of Payne and Cater. This is based upon the argument that the deaths were without motive, evidenced no distinguishing characteristics, and thus could not tend to show plan, scheme, pattern, bent of mind, or identity, the purposes for which the trial court admitted the evidence.

As has already been noted, the test for admissibility of independent crimes in Georgia is that the independent crimes must be similar or logically connected to the crime for which the accused is on trial. *State v. Johnson*, 246 Ga. 654 (272 SE2d 321) (1980). The comparative profile illustrates the necessary similarities and logical connections. Each of the victims was a young black male from a low income home in the metropolitan Atlanta area whose life style placed him alone on the streets; they had a commonality of appearance and behavior. Each was connected to Williams by animal hair and fibers; many of the victims were seen by third parties in the company of Williams, although Williams continually maintained that he did not know and had not associated with any of the victims. Eight of the ten suffered death by asphyxiation; the other two were linked by other characteristics. The actual number of homicides with these common *784 characteristics shows a pattern.

Several witnesses testified to statements made by Williams evidencing his attitude toward those meeting the profile of the victims. There was evidence that he had a negative attitude toward black children from lower class backgrounds. He expressed anger and shame when discussing "street kids" and would make derogatory comments about his own race. He told an acquaintance that he had compiled statistical data on the reduction of the black population by the elimination of one black male. Another young black male who saw victim Geter with Williams related that he had been approached by Williams about a job and got into appellant's car and rode around town with him. He stated that Williams fondled him and that when Williams stopped the car to get something out of the trunk he ran away.

The ultimate issue in determining the admissibility of evidence of other crimes is not mere similarity but relevance to the issues in the trial of the case. The inquiry is whether the proof of the extrinsic crimes tends to prove the crimes charged. This may not be proved by showing the commission of other crimes to prove that the accused has a criminal nature. While one or two additional unexplained homicides which could be associated with Williams might not meet the relevancy test, the sheer number of victims with common characteristics, each logically connected with Williams by hairs and fibers, tends to show a pattern of killings. The murders of Cater and Payne fit this pattern, and the evidence tends to show they were a part of that pattern.

For the independent crimes to be admissible, it was also necessary for the state to offer proof that Williams was the perpetrator of the crimes. In this regard, we have held that the standard of proof of reasonable doubt is not applicable to the proof that the defendant was the perpetrator of the independent crimes, but only to the proof that he or she was the perpetrator of the crimes for which the accused is on trial. *Wallace v. State*, 246 Ga. 738 (273 SE2d 143) (1980). While the evidence that Williams was the perpetrator of the extrinsic offenses was stronger in some cases than others, in each case there was some evidence connecting Williams with the victim, including hair, fibers, blood stains, and eyewitness testimony of Williams associating with some of the victims. We find that this evidence was sufficient to satisfy this prong of the test of admissibility of independent crimes. The court charged the jury on the limited purpose for which the evidence was offered after the presentation of evidence for each victim and again in the final charge at the conclusion of the case. All of the extrinsic crime evidence tended to prove a pattern, and the evidence of appellant's participation *785 in the pattern and involvement with each victim was sufficient to be considered by the jury for the limited purpose for which it was offered.

In conclusion, we find that the circumstances of the other homicides all evidence similarities and connections to each other and to Cater and Payne, and that from the numbers and the patterns it is possible to find a logical connection. Because the extrinsic crimes were relevant to the issues in this case, evidence of them was therefore admissible.

Appellant also argues in this enumeration of error that the state prejudicially injected evidence of "pattern crimes" in violation of the trial court's January 6, 1982 pre-trial order. The pertinent part of that order provides that, before evidence of independent crimes would be admitted, the court would hold a hearing out of the presence of the jury to determine if the requirements for the admission of independent crimes had been met. Appellant argues that the first violation of this order occurred when Dr. Zaki, the medical

examiner who performed the autopsy on Jimmy Payne, testified on redirect examination that one of the reasons he classified Payne's death as a homicide was that several other bodies had been found in rivers around the Atlanta area. After this testimony, appellant moved for a mistrial, which the trial court denied. Appellant contends that Dr. Zaki's testimony constituted prejudicial error, and that the trial court erred in denying his motion for mistrial. We disagree.

The decision whether to grant a motion for mistrial lies within the sound discretion of the trial court, and, moreover, the exercise of that discretion will not be disturbed on appeal unless the grant of a mistrial is essential to preserve the right to a fair trial. *Ladson v. State*, 248 Ga. 470 (12) (285 SE2d 508) (1981); *Spraggins v. State*, 240 Ga. 759 (2) (243 SE2d 20) (1978). In the instant case, an examination of the transcript reveals that, before Dr. Zaki's redirect examination, defense counsel thoroughly cross-examined Dr. Zaki concerning what factors he had used to determine that Payne's death was a homicide, and, in particular, he asked Dr. Zaki if he told a police investigator that, if it had not been for the other killings, he would have ruled Payne's death an accidental drowning. Because defense counsel referred to the other killings, and because he raised the issue of what factors Dr. Zaki had used to determine Payne's death was a homicide, he waived his right to object to Dr. Zaki's reference to these matters on redirect examination. *Halm v. State*, 125 Ga. App. 618 (1) (188 SE2d 434) (1972); see also *Shepherd v. State*, 234 Ga. 75 (1) (214 SE2d 535) (1975). Therefore, the trial court's denial of his motion for mistrial was not error. *Ladson v. State*, *supra*; *Spraggins v. State*, *supra*.

Appellant also argues that the testimony of Lee Brown, the *786 then-Commissioner of Public Safety for the City of Atlanta, violated this pre-trial order. Brown testified as to the organization and responsibilities of a police task force. He testified that task forces are usually organized because they are a proven vehicle for conducting investigations into similar crimes which occur in more than one jurisdiction. He testified that a task force was organized in Atlanta to investigate a pattern of crimes involving the deaths of young black children. He also testified that in April 1981 the task force staked out bridges over rivers in the metropolitan Atlanta area in an attempt to apprehend whoever was responsible for throwing bodies into rivers in the Atlanta area. Appellant objected to the reference to these other crimes and to the use of the word "pattern", but the trial court overruled these objections. On appeal, appellant contends that the injection of this testimony violated the pre-trial order and was prejudicial error. We disagree.

Assuming that Brown's testimony was a violation of the pre-trial order and prejudiced appellant by insinuating that he was responsible for this pattern of other crimes, we find

that such an error is harmless. *Johnson v. State*, 238 Ga. 59, 61 (230 SE2d 869) (1976). First, appellant requested the trial court during voir dire to read a list of all the murdered children to the jury panel and to ask the panel if they knew or were related to any of them. While questioning at least one jury panel the trial court read the list and referred to the children as the "murdered and missing children." Because the attention of the jurors was directed to the missing and murdered children at that early stage of the trial, we find no error. Moreover, Brown did not state the number of pattern deaths to which he referred, and, as we have already determined, evidence of ten independent or "pattern" crimes was properly admitted over a period of several days at trial.

5). Appellant's seventh enumeration of error is based upon his assertion that the court erred in permitting a state's expert witness to discuss mathematical probabilities concerning the fiber evidence and in permitting the prosecutor to argue mathematical probabilities to the jury.

Neither of these contentions has merit, as experts are permitted to give their opinions, based upon their knowledge, including mathematical computations. *Stewart v. State*, 246 Ga. 70, 75 (268 SE2d 906) (1980). Counsel are given wide latitude in closing argument, and are not prohibited from suggesting to the jury inferences which might be drawn from the evidence. *Wisdom v. State*, 234 Ga. 650, 655 (217 SE2d 244) (1975). Such suggestions may include those based upon mathematical probabilities.

6). In his eighth and ninth enumerations of error, appellant argues that the evidence presented is insufficient to support his *787 convictions. *Jackson v. Virginia*, 443 U.S. 307 (99 SC 2781, 61 LE2d 560) (1979); *Strickland v. State*, 250 Ga. 624 (1) (300 SE2d 156) (1983). We disagree. After reviewing the evidence in a light most favorable to the jury's verdict, we conclude that a rational trier of fact could reasonably have found Williams guilty of the murders of Payne and Cater beyond a reasonable doubt. *Jackson v. Virginia*, *supra*; *Strickland v. State*, *supra*.

7). In his tenth and twenty-fifth enumerations of error, and, in part, his thirty-second, Williams claims that he was denied a fair trial because exculpatory information was withheld from him. The record shows that Williams filed several pre-trial motions requesting the court to compel the state to disclose a broad range of exculpatory information. On August 27, 1981 he moved for the court to conduct an in camera inspection of the state's files and to preserve for appellate review all items not disclosed to the defense after the inspection. This motion was granted, and the court subsequently issued orders marking the completion of its review of the files. In the orders the court directed that photocopies of the files pertaining to Payne, Cater, and Williams be sealed and filed with

the clerk of the superior court for appellate review, and, listing twenty-six files it had reviewed in addition to those of Williams, Payne, and Cater, the court stated that photocopies of Brady information from all twenty-nine files had been furnished to defense counsel. During the trial the court added a thirtieth file to the list of files it had reviewed and from which Brady material had been furnished to counsel for the defendant. In addition to the court's in camera inspection, the defendant also had the advantage of a pre-trial offer by the state to look at certain investigatory files concerning Payne and Cater, although it is unclear exactly what material this offer encompassed or to what extent the defense availed itself of this opportunity.

Williams now claims that, despite the above efforts to honor his due process right to a fair trial, various Brady materials were nevertheless withheld from him. For the purpose of evaluating this claim, we have grouped his allegations into five categories. First, he contends that, although the prosecution attempted to prove that fibers and hairs found on the victims were similar to those found in his environment, there is nothing in the Brady files given to the defense which indicates that the state attempted to eliminate the possibilities that the evidence found on the deceaseds' bodies may have actually derived from their environments or from contamination during the recovery of the bodies. Because the testimony concerning these comparisons was crucial to the state's case, Williams contends, any scientific tests which were run on samples obtained from these possible sources, and which did show *788 those fibers or hairs to be similar to those found on the victims, would be material and exculpatory. Moreover, if on the other hand there was inadequate collection and testing of these materials, that fact would also be favorable to the defendant, since it would create a strong doubt concerning the reliability of expert testimony about fiber and hair comparisons. Williams argues that, without being provided access to all information concerning the state's investigations in this area, he has no way to ascertain whether either of the foregoing situations exists, and therefore it was error to withhold this information from him.

Second, he points out that one of the essential elements of the state's case was its attempt to show that the green carpet in his bedroom was exclusively composed of a rare type of fiber, the Wellman 181-b asymmetrical trilobal nylon fiber, which allegedly also was the same type of fiber discovered on many of the victims. Williams says that after his conviction he obtained the services of two fiber experts named Barndt and White, who perused the carpet and arrived at conclusions concerning its nature which were at odds with those of the state's experts. Specifically, he says that affidavits sworn to by Barndt and White show that they found that the carpet also contained common symmetrical trilobal fibers, and that this fact was so obvious that the state must have known it, but suppressed

it. In a related argument, he asserts that the trial court erred by refusing to admit the affidavits of Barndt and White into evidence at the hearing of his motion for new trial.

Third, he says that the prosecution failed to correct false trial testimony by a state's witness. Fourth, he complains of numerous miscellaneous alleged instances of suppression of evidence. Fifth, he criticizes the failure of the trial court to preserve for appellate review the bulk of the files it inspected, and its omission to make a record of the material which it selected from all the files and turned over to trial counsel.

After careful review, we conclude that none of these allegations has any merit. As to the fifth category, we first note that appellant did not object prior to his motion for new trial concerning these aspects of the manner in which the court carried out the in camera inspection. Second, although the record clearly indicates that appellant received and used during trial and post-trial proceedings the Brady material selected by the court, his counsel did not take the elementary step of seeking to make it part of the record. Further, any error in this regard is harmless because, as noted below, appellant has failed to satisfy his burden to show the materiality and favorable nature of the evidence sought, thus forestalling any need to call for the material reviewed by the trial court. See *Welch v. State*, 251 Ga. 197 (8) (304 SE2d 391) *789 (1983).

The allegations comprising the fourth category are also nonmeritorious. Many are unsupported by argument or citation of authority, and must be deemed abandoned. *Rucker v. State*, 250 Ga. 371 (4) (297 SE2d 481) (1982). Moreover, many lack any citation to the record and transcript of this case, and others make reference to documents and incidents which are not part of the record. "On appeal, this court can only consider and correct errors which appear in the record." *Dean v. State*, 250 Ga. 77 (2) (b) (295 SE2d 306) (1982). In instances where we have been able to ascertain the sections of the record and transcript which are pertinent to these allegations, we have reviewed them to determine whether they have any merit, but have found that they share the same deficiencies as the first, second, and third groups of allegations. We shall now address those deficiencies.

"[T]he appellant has the burden of showing he was denied material exculpatory information such that he was denied a fair trial." *Kilgore v. State*, 251 Ga. 291 (5) (305 SE2d 82) (1983). If the trial court performs an in camera inspection and denies the defendant access to certain information, on appeal the appellant has the burden of showing both the materiality and the favorable nature of the evidence sought. *Welch v. State*, supra. Mere speculation that the items the appellant wishes to review possibly contain exculpatory information does not satisfy this burden. *Id.* "If the appellant desires to have this inspection reviewed by this court, she must point out what material she believes to have

been suppressed and show how she has been prejudiced." *Ledesma v. State*, 251 Ga. 487 (10) (306 SE2d 629) (1983). "Brady does not impose an affirmative duty on the prosecution to seek out information for the defense, even if such information is more accessible to the prosecution than to the defense." *Hines v. State*, 249 Ga. 257 (1) (290 SE2d 911) (1982). And, there is no suppression of evidence by the prosecution if it is not part of the state's files, and its existence is not known to the prosecutor or to any law enforcement officer. *Brown v. State*, 250 Ga. 66 (6) (295 SE2d 727) (1982).

With respect to the second category of appellant's allegations, the Barndt and White affidavits were among the materials forwarded on appeal, and, despite the trial court's refusal to admit them into evidence at the hearing of the motion for new trial, we have nevertheless reviewed them for the purpose of this Division. See Division 19, *infra*. However, even taking them into account, we find that, as with the first, third, and fourth categories, appellant has clearly failed to carry his burden, since he relies upon speculation, rather than proof, that the alleged exculpatory evidence exists; that its existence was known to the prosecution prior to or during his trial; and that it was part of the state's files. We therefore find no error.

8) a). In enumeration eleven, appellant contends that the trial court erred when it denied his September 10, 1981 motion to suppress. In that motion, Williams sought to suppress statements he gave to the police and observations they made in the course of his detention during the early morning hours of May 22, 1981. Following a hearing on this motion to suppress, the court entered an order denying it, in which the court found that the initial stop of appellant was justified under *Terry v. Ohio*, 392 U.S. 1 (88 SC 1868, 20 LE2d 889) (1968), and that appellant's consent to the ensuing detention and search of his automobile was given voluntarily and was not the result of duress or coercion.

The transcripts of testimony at the September 10, 1981 pre-trial hearing and at trial show that prior to May 22 the bodies of several young black males had appeared in local rivers, and that the police, speculating that these corpses had been thrown into the water from bridges, and that the same might occur with other victims, had stationed stakeout teams at several bridges over the Chattahoochee River. On the evening of May 22 a four-man surveillance team was stationed at the James Jackson Parkway Bridge, which is a two-lane bridge crossing the Chattahoochee from the northwest to the southeast, with the northwesterly end connecting with South Cobb Drive in Cobb County, and the southeasterly end feeding into the James Jackson Parkway in Fulton County. Near the western terminus of the bridge is an entrance ramp from South Cobb Drive onto Interstate 285.

The stakeout team consisted of FBI Agent Gilliland, Atlanta police officer Holden, and two Atlanta police recruits, Jacobs and Campbell. Holden and Gilliland, who were dressed in plain clothes, were stationed in concealed unmarked chase cars at each end of the bridge, with Campbell and Jacobs positioned on foot on the west bank of the river and at the southeastern corner of the bridge, respectively. The location of the team members permitted observation of the reflection of lights from the headlamps of cars approaching from the west before the cars themselves reached the bridge. In addition, cars entering the bridge from the west and travelling ten miles per hour or more could be heard by Campbell to make a distinctive noise as they passed over a metal expansion joint on the bridge. The members of the team operated under orders to maintain surveillance of the bridge from the hours of 8:00 p. m. to 6:00 a. m.

Early that morning a loud splash was heard, which, according to Campbell, sounded like the sound of a human body hitting the water *791 below the bridge. No lights had been observed up to that point, nor had the characteristic noise of the expansion joint been heard. There was testimony that the vehicular traffic was light at that hour of the day, and that a period of at least ten minutes elapsed between the time the last car was seen to cross the bridge and the sound of the splash. Shortly after the splash, a car's lights appeared on the bridge directly above where the splash had occurred, and were seen to start moving slowly toward the Fulton end of the bridge. The car was unusually close to the southern edge of the bridge, and was travelling at approximately three to four miles per hour. It was observed to exit the bridge at the Fulton end, turn around without stopping in a nearby parking lot, and proceed back across to Cobb County at about thirty-five to forty miles per hour. No other autos passed over the bridge during this incident.

Gilliland and Holden followed the car and saw it enter Interstate 285 from the South Cobb Drive ramp and begin travelling fifty to fifty-five miles per hour. They stopped the car on the expressway and, during the ensuing sixty to ninety minutes, the driver, who proved to be Wayne Williams, was interrogated by several different officers and agents. His vehicle was also searched. Agent Gilliland testified at trial that after he and Holden stopped Williams, Gilliland asked Williams whether it was all right if they looked in his car, and that Williams told him they could. During his own trial testimony, Williams stated that this aspect of Gilliland's testimony was truthful. Moreover, during cross-examination he was asked the following question: "Now, when you were stopped the morning hours of May 22nd, you readily and voluntarily talked to the officers there, did you not?", to which he replied, "I sure did."

On appeal Williams, while conceding that the initial stop was permissible as a brief investigatory detention under *Terry v. Ohio*, supra, contends that his detention exceeded the scope of a permissible *Terry* stop and became a full-blown arrest. This arrest was illegal, he contends, because it was not supported by probable cause, and he argues that all statements, observations, and fruits of that illegal detention (including the search warrants authorizing June 3 and June 22 searches of his home and automobile) were unauthorized and should have been suppressed. The state counters by arguing that the initial detention and questioning of appellant were supported by "articulable suspicion" as required by *Terry v. Ohio*, supra, and that appellant consented to his continued detention and to the search of his automobile.

i). First, we address the correctness of the trial court's ruling that appellant's consent to his continued detention and to the search of his vehicle was given voluntarily. The applicable general principles *792 are well known. Where the state relies on consent to justify a warrantless search, it "has the burden of proving that the consent was, in fact, freely and voluntarily given." *Bumper v. North Carolina*, 391 U.S. 543, 548 (88 SC 1788, 20 LE2d 797) (1968). A valid consent eliminates the need for either probable cause or a search warrant. *Dean v. State*, 250 Ga. 77, 80 (295 SE2d 306) (1982); *Hall v. State*, 239 Ga. 832 (238 SE2d 912) (1977).

We have examined the record of the September 10 suppression hearing and the portions of the trial transcript relating to this issue, and are convinced that appellant's consent, when considered in the totality of the circumstances surrounding the May 22 stop, was given freely and voluntarily. The critical factor is that appellant's own testimony establishes that he consented to being questioned and having his car searched. Unless clearly erroneous, the trial court's rulings on disputed facts and credibility at a suppression hearing must be accepted by this court on appeal, *Dean v. State*, supra; *Woodruff v. State*, supra, and because the court's finding in the instant case on the consent issue is supported by the evidence, we affirm.

ii). Appellant further contends that, regardless of the legality of the initial Interstate 285 stop and of any consent he may have subsequently given, his prolonged detention by police exceeded the scope of a *Terry*-type stop and became a full-fledged arrest which was not supported by probable cause, and was thus illegal. We cannot agree.

The rationale of *Terry v. Ohio*, supra, and its progeny allows investigating officers who possess articulable suspicion of criminal activity to detain a suspect for a limited period in order to identify the suspect, frisk him if necessary, and conduct limited questioning. See *Radowick v. State*, 145 Ga. App. 231, 237 (244 SE2d 346) (1978). Detention beyond that

authorized by Terry is an arrest, and, to be constitutional, such an arrest must be supported by probable cause. *United States v. Berry*, 670 F2d 583 (5th Cir. 1982). Probable cause to arrest exists where, based on objective facts and circumstances, a man of reasonable caution would believe that a crime has been or is being committed. *Brinegar v. United States*, 338 U.S. 160, 175-76 (69 SC 1302, 93 LE 1879) (1949).

Our review of the record shows that appellant was never formally arrested by police on May 22; however, it is not entirely clear from the record before us whether appellant was "arrested" in the sense contemplated by the Fourth Amendment and by decisions such as *United States v. Berry*, *supra*. But, we need not belabor this point because we hold that even if a *de facto* arrest occurred, the officers at the scene of the May 22 stop possessed probable cause to detain *793 appellant and question him thoroughly concerning his presence on the Jackson Parkway Bridge. The facts and circumstances known by the officers, including the recent dumping of the body of Jimmy Ray Payne in the area, the silent and unseen approach of appellant's car to the bridge, and the loud splash heard by recruit Campbell at the same time as the car appeared to have been stopped on the bridge, warranted, in a man of a reasonable caution, the belief that a crime had been committed by appellant. Cf. *Bothwell v. State*, 250 Ga. 573 (300 SE2d 126) (1983). We find no error.

iii). In his appellate briefs and arguments, appellant contends that any statements he made on the night of May 22 were improperly admitted into evidence because he was never advised by his questioners of his right to remain silent under *Miranda v. Arizona*, 384 U.S. 436 (86 SC 1602, 16 LE2d 694) (1966). However, we need not address the merits of this contention, because appellant failed to adequately raise this issue or invoke a ruling by the trial judge concerning the admissibility of his statements under *Miranda*, either before or during trial. To the contrary, counsel for appellant himself first injected much of the substance of these statements into the trial during his opening argument. Having failed to object to the admissibility of the contested evidence at trial, appellant cannot now raise the issue for the first time on appeal. See *Lambert v. Jones*, 250 Ga. 603 (299 SE2d 716) (1983); *Allen v. State*, 233 Ga. 200, 201 (210 SE2d 680) (1974).

b). In enumeration twelve appellant cites as error the trial court's denial of his motion to suppress evidence seized from his house and automobile in two separate searches, conducted June 3 and June 22, 1981. For the following reasons, we hold that the warrants were supported by the requisite probable cause, and affirm.

On June 3, 1981 Officer Morris Redding, Chief of the Task Force on Missing and Murdered Children, appeared before a superior court judge to request a warrant authorizing a search of appellant's residence and automobile. In this connection Redding executed an affidavit

containing 157 paragraphs, which provided an exhaustive listing of the facts and circumstances, and which demonstrated, according to Redding, that evidence of the crime of murder was located in appellant's house and car. The following facts and circumstances were included in the affidavit: a description of seventeen Task Force victims, the cause of their deaths, and the fact that state crime lab analysis had revealed the presence of microscopic fibers or dog hairs, or both, on each of the bodies; the circumstances of the May 22 Jackson Parkway Bridge stop of appellant, including the search of appellant's car and observations of dog hairs, a nylon cord, and a paper bag containing men's clothes; the subsequent police *794 investigation leading to the alleged discovery that appellant gave false information to the officers who questioned him on May 22; and the details of police surveillance of appellant which was conducted subsequent to the May 22 stop. The warrant was issued on June 3, and a search of appellant's residence and car was conducted that day.

On June 22 Redding applied for a second warrant with an affidavit which was identical to the earlier affidavit, except that twelve new paragraphs describing the June 3 search were appended to the initial 157 paragraphs. The additional paragraphs stated that certain fibers recovered in that search were determined by scientific tests to be "microscopically identical" to fibers found on the bodies of sixteen of the Task Force victims, and requested that another search be allowed to gather evidence to corroborate the results of scientific tests already performed. The warrant was issued, and a search was conducted June 22. At trial evidence gathered from appellant's house and car, especially fiber evidence, was used extensively to support the state's case.

i). On appeal Williams contends that, first, several of the recitals in the affidavits were based on observations made by police or statements given by appellant during the May 22, 1981 expressway stop of appellant. These portions of the affidavit, he contends, are impermissibly tainted by illegal police activity, should be excluded as "fruit of the poisonous tree," *Wong Sun v. United States*, 371 U.S. 471 (83 SC 407, 9 LE2d 441) (1963), and cannot be used to supply probable cause for the later searches of his residence and car. However, as we have already concluded that appellant consented to the May 22 search, see subdivision (a) (i), *supra*, and that his detention was supported by probable cause to arrest, see subdivision (a) (ii), *supra*, we hold that his "fruit of the poisonous tree argument" is meritless.

ii). Appellant also attacks the facial validity of the affidavits on several grounds. He first contends that the affidavits listed numerous items (including fibers from carpets, cords, blankets, and upholstery; dog hairs; firearms; knives; photographs; and various documents) which police expected to find upon a search of appellant's automobile and

apartment, without providing any facts or circumstances to warrant a man of reasonable caution to believe that the articles sought were actually located in the appellant's home or car.

We note at the outset that "[w]hen an application for a search warrant has been made by the police to a neutral and detached magistrate, and the magistrate has issued the warrant based on a finding of probable cause, a reviewing court will pay substantial deference to the magistrate's finding." *Devier v. State*, 247 Ga. 635, 637 (277 SE2d 729) (1981). In *Murphy v. State*, 238 Ga. 725 (234 SE2d *795 911) (1977), we stated that before a warrant may issue, the issuing magistrate must have sufficient reasons to believe that a crime was committed, that the items sought are connected with the crime, and that the items sought will be found in the place to be searched. *Id.* at 727. We held there that probable cause existed for the issuance of a warrant where a police officer, in the supporting affidavit, stated that he "presumed" that a gun and prying tool, which had been used in a robbery the night before, were located at the suspect's residence or car on the following day, and we recognized that under certain circumstances an affiant may reasonably infer that items connected with a crime will be found in a specific location.

Here, on the night of May 22 police officers observed that appellant's car contained various suspicious articles, including a length of nylon rope, a paper bag which appeared to contain men's clothing, and loose dog hairs. On May 24 the body of Nathaniel Cater, who was killed by asphyxiation, was found floating in the Chattahoochee River downstream from where appellant had been stopped and questioned. An examination of Cater's body revealed the presence of dog hairs. It was allegedly discovered in the course of a police investigation that statements made by appellant concerning his presence at the Jackson Parkway Bridge that night were fabrications. Given these facts, a man of reasonable caution could, we think, infer that evidence of a crime would be discovered in either appellant's automobile or residence. See *State v. Babb*, 134 Ga. App. 302 (214 SE2d 397) (1975); *Driscoll v. State*, 129 Ga. App. 702 (201 SE2d 11) (1973).

Appellant also challenges Redding's affidavits on the ground that the information recited therein was not within Redding's personal knowledge and did not establish the credibility of the various sources relied on by the affiant, including fellow police officers, FBI agents, and crime lab analysts. This contention is without merit. The rule in Georgia is that, in the context of an application for a search warrant, the veracity of information received from government officials and the reliability of those officials may be assumed. *Devier v. State*, *supra*, at 638. "Observations by fellow officers of the Government engaged in a common

investigation are plainly a reliable basis for a warrant applied for by one of their number." *United States v. Ventresca*, 380 U.S. 102, 111 (85 SC 741, 13 LE2d 684) (1965).

Finally, appellant contends that the supporting affidavits of Officer Redding contained misrepresentations of fact and false statements, to the extent that the warrant was void. See *Franks v. Delaware*, 438 U.S. 154 (98 SC 2674, 57 LE2d 667) (1978). In the *Franks* case, the Supreme Court set out the following procedure for *796 challenging an affidavit on these grounds:

"[W]here the defendant makes a substantial preliminary showing that a false statement knowingly and intentionally, or with reckless disregard for the truth, was included by the affiant in the warrant affidavit, and if the allegedly false statement is necessary to the finding of probable cause, the Fourth Amendment requires that a hearing be held at the defendant's request." *Id.* at 155-56.

On November 20, 1981, the trial judge held a pre-trial hearing on appellant's misrepresentation claims, and denied the motion to suppress, concluding that appellant failed to shoulder his initial burden of making a "substantial preliminary showing" that a false statement was made by Redding in his affidavit. The only evidence introduced by appellant on this point concerned (1) a slight discrepancy between the testimony of recruit Jacobs at the preliminary hearing and Redding's description in the affidavit of the Jackson Parkway Bridge incident; and (2) the fact that, in the June 22 affidavit, Redding claimed that the fibers found in the June 3 search of appellant's bedroom and those found on Cater's body were "microscopically identical," but that this precise characterization of the similarity between those fibers could not be attributed to any of the experts who had supplied information to Redding. We agree with the trial court that these allegations wholly fail to raise an issue of intentional or reckless falsehood under *Franks v. Delaware*, *supra*. See *Nutter v. State*, 162 Ga. App. 349 (291 SE2d 423) (1982). We also note that the court ruled at the hearing that, even disregarding the alleged misrepresentations, the Redding affidavits were sufficient to establish probable cause. This cured any possible error. See *Franks v. Delaware*, *supra* at 155-56; *Waller v. State*, 251 Ga. 125 (303 SE2d 437) (1983). This enumeration is without merit.

9). Williams contends that the trial court erred in quashing subpoenas served upon George Napper and B. L. Neikirk of the Atlanta Bureau of Police Services.

a). The subpoena to Neikirk sought production of the following information: "statistical information contained in notes, books, records, writings and/or documents pertaining to any and all homicides both solved and unsolved, either committed within the city limits of

Atlanta, Fulton County or the surrounding counties in which bodies were found from January 1972 to the present." This subpoena is clearly overbroad. The requested information spans a period of almost a decade and pertains not only to the City of Atlanta and Fulton County, but also to an undefined number of "surrounding counties," rendering it both "unreasonable and oppressive." See OCGA 24-10-22 (b) (Code Ann. 38-801). There was no error here.

b). The subpoena to Napper requested production of "[r]ecords *797 relating to any case involving [N]egro children between the ages of 6 and 28 years of age, who have been reported missing or who have been found dead from January 1, 1981 to present. OR a list compiled from the above records giving the name, age, physical description, dates, parents' names, suspects arrested, and evidence involved in any APD investigation of the above." As the information sought did not bear upon appellant's guilt or innocence of the offenses charged in the indictment, this enumeration is without merit.

c). As for enumerations fourteen and sixteen, the trial court did not err in quashing the subpoenas served upon Governor Busbee, Attorney General Bolton, GBI Director Peters, Executive Counsel Tidwell, Acting U. S. Attorney Kirkley and FBI Agent in Charge Glover. The information sought pursuant to these subpoenas concerned supposed discussions which took place at the Governor's Mansion regarding the Williams case prior to his indictment — all of which are irrelevant to the present case.

10). In his seventeenth enumeration of error, appellant argues that he was denied a fair trial because the sequestered jurors were exposed to a drawing, which someone had placed on the wall of their dining room, depicting the outline of a man's face with the word guilty appearing above it.

However, appellant waived his right to raise this issue on appeal. First, when the jury's exposure to the drawing was disclosed to defense counsel, he made no motion for a mistrial. *Martin v. State*, 242 Ga. 699 (251 SE2d 240) (1978); *Thomas v. State*, 158 Ga. App. 97 (2) (279 SE2d 335) (1981). Second, the day after the incident occurred, the trial court first described to defense counsel the precautions he had ordered to insure that no other such incidents would take place, and then asked him if he wanted "to say anything as relates to that." Defense counsel responded that he had nothing further. Moreover, even if a motion for mistrial had been made, its denial would not have been an abuse of discretion, as we find that the incident was not so prejudicial as to prevent the defendant from receiving a fair trial. *Messer v. State*, 247 Ga. 316 (6) (276 SE2d 15) (1981).

11). In his eighteenth enumeration of error, appellant argues that the trial court denied him both the right to be present at all stages of the trial and the right to a fair trial by making prejudicial remarks to the jury outside of his presence. The remarks of which appellant complains followed the end of testimony on February 5, 1981. When court recessed, the trial judge stated, in the presence of counsel and appellant, that he was going to meet with the jurors before they left. Immediately thereafter, appellant's counsel addressed the court, stating that it was appellant's desire that married jurors be allowed a conjugal visit for the upcoming weekend. Defense counsel specifically *798 asked appellant if that was his desire, and appellant answered affirmatively. The district attorney agreed to this request, and the trial court immediately proceeded to inform the jury of this information out of counsel's and appellant's presence. Appellant argues that, when the trial court addressed the jury, it improperly implied that defense counsel played the "heavy" by not agreeing to allow unmarried jurors to have visitors over the weekend. Moreover, appellant argues that on that occasion the trial court also impermissibly pressured the jury by emphasizing the importance of the trial and implying that, if they acquitted a mass murderer, they would have to justify their action to the rest of the community. We disagree with appellant's arguments.

First, although the accused has the right to be present at all stages of his or her trial, *Gilreath v. State*, 247 Ga. 814, 824 (3) (279 SE2d 650) (1981), cert. denied 456 U.S. 984 (102 SC 2258, 72 LE2d 862); *Wilson v. State*, 212 Ga. 73, 74 (90 SE2d 557) (1955), this right can be waived if the accused's counsel makes the waiver in the accused's presence, or with his or her express authority, or if the accused subsequently acquiesces to the waiver made by counsel. *State v. Phillips*, 247 Ga. 246 (1) (B) (275 SE2d 323) (1981); *Wilson v. State*, supra, at 77-78. In the instant case, assuming without deciding that appellant had the right to be present at the discussion between the court and jury, we find that appellant waived this right because it appears that appellant's counsel, with appellant's understanding and acquiescence, implicitly agreed for the trial court to discuss the matter of conjugal visits with the jury out of appellant's presence. *State v. Phillips*, supra, p. 248; *Wilson v. State*, supra, pp. 77-78; *Bailey v. State*, 249 Ga. 535 (6) (291 SE2d 704) (1982). This conclusion accords with appellant's statement on appeal that the defense had an agreement with the trial court waiving the defense's presence while the court discussed the matter of the conjugal visit with the jurors. Moreover, although the scope of the trial court's statements may have exceeded the initial purpose of its discussion with the jury, they were not so prejudicial as to deny appellant a fair trial. The trial court did not single out defense counsel as the party responsible for the limitation of visitors to married jurors; rather, it informed the jury that both the state and the defense were responsible for the agreement. See, *Lahr v. State*, 239 Ga. 813 (3) (238 SE2d 878) (1977); *Swint v. State*, 203 Ga. 430 (6)

(47 SE2d 65) (1948). Furthermore, the transcript shows that the trial court did not unduly place pressure on the jury.

For the above reasons, we find that the trial court's actions did not deny the defendant a fair trial or the right to be present at all stages of the proceedings.

*799 12). In his nineteenth enumeration of error, Williams argues that he was denied a fair trial because the trial court improperly limited his cross-examination of a juvenile witness.

On cross, defense counsel asked the witness three questions for impeachment purposes: first, how many times he had been arrested for stealing; second, how many times he had been convicted of stealing; and third, if he stole quite frequently. The state's objections to each of these questions were sustained by the trial court.

On appeal, appellant first argues that the trial court improperly limited his cross-examination of this witness by ruling that he could not use the witness' juvenile record for purposes of impeachment. *Davis v. Alaska*, 415 U.S. 308 (94 SC 1105, 39 LE2d 347) (1974). However, contrary to appellant's characterization of the court's ruling, our inspection of the transcript shows that the trial court did not rule that the defense could not use the witness' juvenile record for impeachment. Rather, although the prosecution made assertions in that vein, the trial court's only rulings were to the effect that the defense could not impeach the witness' character by merely questioning him concerning his convictions or proclivity for stealing. We find that the court's rulings were correct. Instances of specific misconduct may not be used to impeach a witness' character or veracity unless the misconduct has resulted in the conviction of a crime involving moral turpitude, OCGA 24-9-84 (Code Ann. 38-1804); *Green*, Ga. Law of Evidence (2nd ed.), 138; *Daniel*, Ga. Crim. Trial Prac. (2nd ed.), 20-23; *Lewis v. State*, 243 Ga. 443 (254 SE2d 830) (1979); *McCarty v. State*, 139 Ga. App. 101 (1) (227 SE2d 898) (1976), and the proper method of proving such a conviction is by the introduction of a certified copy thereof. *Daniels v. State*, 234 Ga. 523 (3) (216 SE2d 819) (1975). Nevertheless, if no best evidence objection is interposed, a witness' answer to the effect that he or she has been convicted of a crime is admissible to prove the crime, *Moret v. State*, 246 Ga. 5 (3) (268 SE2d 635) (1980); however, in the instant case the state did object on that ground. Moreover, despite the trial court's intimations to defense counsel that he would be allowed to make further attempts to impeach the witness, the record shows that the defense never attempted to impeach him with his juvenile record. For these reasons, appellant's argument that the trial court improperly prohibited him from using the witness' juvenile record for purposes of impeachment is without merit, and we thus do not need to reach appellant's contentions regarding *Davis v. Alaska*, *supra*. See generally *Favors v. State*, 234 Ga. 80 (3) (214 SE2d

645) (1975); *Gilstrap v. State*, 250 Ga. 814 (2) (301 SE2d 277) (1983); *Smith v. State*, 154 Ga. App. 190 (3) (267 SE2d 826) (1980).

Appellant also appears to be arguing in this enumeration that, *800 even if the trial court correctly ruled that he could not introduce the witness' juvenile record, it erred in not allowing him to question the witness about his juvenile record to uncover any bias or interest that may have slanted the witness' testimony. See *Hines v. State*, 249 Ga. 257 (2) (290 SE2d 911) (1982); *Owens v. State*, 251 Ga. 313 (1) (305 SE2d 102) (1983). Again, however, we find appellant's construction of the transcript to be erroneous. The record shows that appellant's trial counsel did not propound any questions concerning the witness' bias or interest, but that he, instead, asked questions relating to the witness' character or trustworthiness: specifically, how many times the witness had been convicted and arrested for stealing and if he stole quite frequently. See *Favors v. State*, 234 Ga., supra, at 87; 3A *Wigmore, Evidence*, 926, p. 750 (Chadbourn Rev. 1970). Moreover, although appellant urges on appeal that bias could have been shown by the fact that the witness was serving a sentence at the time he testified, the existence of this alleged sentence was not made known at trial. See *Wigmore*, 949, p. 790 for a listing of circumstances from which bias could be inferred. Thus, the record shows that the trial court did not improperly limit appellant's cross-examination concerning the witness' bias or interest.

For the above reasons, this enumeration of error is without merit.

13). In an attempt to discredit Bobby Henry's testimony that he had observed appellant and Nathaniel Cater holding hands three days before Cater's body was found, the defense called John Henley as a witness. After establishing the witness knew Henry, counsel inquired, "Does Bobby Henry always tell the truth?" The witness replied, "I can't even answer that ... I can't say yes; I can't say no." Defense counsel then attempted to question the witness about a prior inconsistent statement he had made regarding Henry's veracity. Following the state's objection the trial court called a bench conference. At this time defense counsel stated that the witness had previously told him "Henry was a bald-face liar," and that he wished to impeach the witness with this statement. The trial court ruled that appellant could not impeach his own witness.

In enumeration of error twenty-one, appellant argues this ruling was error, citing *Gibbons v. State*, 248 Ga. 858 (286 SE2d 717) (1982), *Davis v. State*, 249 Ga. 309 (290 SE2d 273) (1982), and *Ranger v. State*, 249 Ga. 315 (290 SE2d 63) (1982). These three cases were decided after the case at hand was tried. Read together these cases stand for the proposition that a party may use a prior inconsistent statement made by his own witness

both to impeach the witness and as substantive evidence without showing surprise. Cf. OCGA 24-9-81 (Code Ann. 38-1801). This rule is not applicable, however, *801 where, as here, the prior inconsistent statement is not admissible in evidence.

OCGA 24-9-84 (Code Ann. 38-1804) supplies the procedure for questioning a witness about his knowledge of the general reputation in the community of a second witness, and, as a corollary, whether, based on that reputation, he would believe the witness under oath. It is clear, however, that only evidence of general reputation is admissible. A witness may not express his private opinion of another's capacity for truthfulness or otherwise comment on his reputation for veracity. *Gordon v. Gilmore*, 141 Ga. 347 (80 SE 1007) (1904); *Taylor v. State*, 17 Ga. App. 787 (88 SE 696) (1916); Agnor, *Georgia Evidence*, 5-7, p. 60.

Appellant's question, "Does Bobby Henry always tell the truth," was, therefore, improper, and the answer appellant sought was not admissible. The question, however, was not objected to. While the trial court's ruling that appellant could not impeach his own witness with the witness' prior inconsistent statement regarding Henry's veracity was based on a principle which has since been substantially altered by this court, the correct result was reached. We find no error.

14). In his twenty-second enumeration of error appellant contends that the trial judge erred by allowing the state to make prejudicial statements to the jury during arguments. For example, he claims that the court was wrong to permit prosecutors to imply that Williams was a mad dog killer and to compare him to Attila the Hun, Adolf Hitler, and Idi Amin. However, appellate counsel for the state respond that Williams failed to object during arguments on any of the grounds he now enumerates as error.

"[I]t has been held that, unless the court's attention is called to such improper argument and a ruling invoked upon the trial, it is too late to raise the point for the first time in a motion for new trial." *Morris v. State*, 200 Ga. 471, 480 (37 SE2d 345) (1946). "When an improper argument is made, whether substantive or procedural, opposing counsel has a duty to act by interposing an objection." *Scott v. State*, 243 Ga. 233 (2) (253 SE2d 698) (1979). Appellant concedes that his attorneys made no objection during trial, but he argues that, for several reasons, this court should nevertheless review his enumeration. First, he claims that under former Ga. Code Ann. 81-1009 (now codified with editorial changes as OCGA 17-8-75), the court has an independent duty to interpose and prevent prejudicial statements even absent timely objection. However, this contention has been decided adversely to Williams. *Hudson v. State*, 250 Ga. 479 (4) (299 SE2d 531) (1983).

In addition, Williams contends that, under the circumstances of *802 this case, the judge had an independent duty to prevent prejudicial statements because appellant's lead counsel, Alvin Binder, was a member of the Mississippi bar and was unfamiliar with the Georgia rules of evidence and criminal procedure. However, there is nothing in the record to indicate the degree of Mr. Binder's familiarity or lack thereof with these rules, and we are thus unable to consider this allegation. Moreover, pretermittting any other critique of appellant's argument, the record shows that he was represented at trial by a total of at least five attorneys, including three members of the Georgia bar. Since he has not demonstrated how the alleged unawareness of one of his foreign attorneys operated to divest the local members of his defense team of their knowledge of Georgia law, this contention has no merit. Williams advances other arguments as to why this court should review his twenty-second enumeration, but they are not persuasive, and we therefore will not entertain his claim of error.

15). In three related enumerations appellant complains that the topic of lie detector tests was referred to in the presence of the jury. In one of the enumerations of error, the twenty-fourth, appellant argues that he was denied a fair trial because the prosecution, on redirect examination of a witness whose testimony was part of the state's proof inculcating Williams in the death of Larry Rogers, asked the witness if he had been given a lie detector test. The witness, Tilbert Baynham, responded affirmatively.

Appellant's trial counsel had already elicited the same testimony from Baynham on cross-examination, and, moreover interposed no objection to the district attorney's question on redirect, or to Baynham's answer thereto. For these reasons, appellant has lost his right to raise this ground on appeal. *Fleming v. State*, 243 Ga. 120 (6) (252 SE2d 609) (1979).

In a related enumeration of error, number twenty-three, appellant argues that the trial court erred in denying a motion for mistrial after Assistant District Attorney Mallard asked witness Robert Henry whether he had been given a lie detector test, and in another enumeration, number twenty-nine, Williams contends that, although the trial court gave curative instructions after Mallard posed the above question, those instructions were insufficient because they were not accompanied by a rebuke to the district attorney.

The transcript shows that Robert Henry testified that he had seen Williams and Nathaniel Cater holding hands outside the Rialto Theatre on Thursday night, May 21, 1981, which was three days before Cater's body was pulled from the river. During subsequent direct examination of Henry the following colloquy occurred:

*803 "[MR. MALLARD]: And, how is it that you're down here today?

"A. Well, the detective found me, you know. He subpoenaed me to come. And he asked me what I knew about it. So I got it off my chest. I told him.

"Q. Until the detective found you, you had not contacted the detective himself or the D.A.'s office, have [sic] you?

"A. No, I haven't. [sic]

"Q. Was that Detective Sidney Dorsey that you saw about it this week?

"A. Yes.

"Q. All right. Did Detective Dorsey take you out and put you on a lie detector?

"MR. BINDER: We object.

"THE COURT: Disregard that. Next question. I instruct the jury to disregard that remark.

"MR. BINDER: I still want to move for a mistrial because he knows better than to ask that question.

"THE COURT: The motion for mistrial is denied."

We agree with appellant that Mallard's question about a lie detector was clearly impermissible, as the parties had not stipulated that the results of the polygraph test administered to Henry would be admissible. Such a stipulation is a prerequisite for the admissibility of such evidence. *State v. Chambers*, 240 Ga. 76 (239 SE2d 324) (1977). Thus, we are faced with the issue whether an improper reference to the fact that a polygraph test has been given warrants the grant of a mistrial.

This court and our Court of Appeals have held that a mistrial is not required on every occasion where a jury has been improperly apprised that a lie detector test has been given. *Porter v. State*, 237 Ga. 580 (3) (229 SE2d 384) (1976); *Roberts v. State*, 243 Ga. 604 (2) (255 SE2d 689) (1979); *Drake v. State*, 142 Ga. App. 14 (1) (234 SE2d 825) (1977); *Snell v. State*, 160 Ga. App. 74 (1) (286 SE2d 52) (1981). Instead, the proper rule is that the decision whether to grant a mistrial is within the sound discretion of the trial court. Moreover, the exercise of that discretion will not be disturbed on appeal unless the grant of a mistrial is essential to preserve the right to a fair trial. *Snell v. State*, supra; *Roberts v. State*, supra; *Ladson v. State*, 248 Ga. 470 (12) (285 SE2d 508) (1981); *Spraggins v. State*, 240 Ga. 759 (2) (243 SE2d 20) (1978); *Lynch v. State*, 234 Ga. 446, 448 (216 SE2d 307) (1975).

In the instant case, the fact that a lie detector test may have been administered to Henry was inappropriately brought to the jury's attention by Mallard. Compare *Snell*, supra, at 75 (fact of lie detector test was divulged by witness in unresponsive answer). Moreover, *804 although Henry did not answer the question, the question clearly prejudiced Williams to some extent, because reasonable jurors could have inferred from it that Henry took and passed such a test. These inferences, if drawn by the jury, would have strengthened the credibility of Henry, whose testimony, if believed, was a key strand in the web of circumstantial evidence against Williams. Nevertheless, although the question suggested that a test had been administered, it did not directly state either that a test had been given, or that Henry had passed it. *Id.* In addition, even though the trial judge did not admonish the prosecutor, he did give prompt curative instructions.

We strongly disapprove of the action of the district attorney; however, under the circumstances of this case ÔÇö a question which only implied that the witness took and passed the test, and prompt curative instructions ÔÇö "we cannot say that the trial court abused its discretion in denying defendant's motion for mistrial or that the defendant was deprived of his right to a fair trial." *Ladson*, supra, at 478. Accord, *Roberts v. State*, supra; *Snell v. State*, supra; *Spraggins v. State*, supra.

16). In enumeration of error twenty-six, the appellant asserts error in the refusal of the trial court to grant his motion to sever the murders of Nathaniel Cater and Jimmy Ray Payne. In so arguing, he relies on *Dingler v. State*, 134 Ga. App. 223, 224 (214 SE2d 6) (1975), which was decided by the Court of Appeals after certifying the question to this court.

In *Dingler v. State*, 233 Ga. 462, 464 (211 SE2d 752) (1975), we adopted the ABA Standards on Joinder and Severance, ABA Standards Relating to the Administration of Criminal Justice, pp. 281-293 (1974). Under these standards, we held that severance upon the request of the defendant is mandatory "where the offenses have been joined solely on the ground that they are of the same or similar character." We also quoted, however, the ABA Standards on Joinder providing: "'Two or more offenses may be joined in one charge, with each offense stated in a separate count, when the offenses, whether felonies or misdemeanors or both: ... (b) are based on the same conduct or on a series of acts connected together or constituting parts of a single scheme or plan.'" (Emphasis supplied.) Under the rules on severance, offenses joined under (b) above, are severable in the discretion of the trial court. *Jackson v. State*, 249 Ga. 751, 758 (295 SE2d 53) (1982).

We have held in *Division 4*, supra, that the evidence supports a finding that the murders were carried out as "parts of a pattern" and that the similar crimes, along with the fiber evidence, were admissible against the appellant on each count. This being so, we do not

find that the trial court abused its discretion in allowing the state to proceed to *805 trial on both counts or erred in overruling the appellant's motion to sever.

17). In his twenty-seventh enumeration of error, appellant contends that the trial court failed "to safeguard appellant's presumption of innocence before an impartial jury...." His argument breaks this enumeration into two distinct complaints: (1) that the trial court erred in not striking certain jurors for cause and in impermissibly limiting voir dire; and (2) that the trial court erred in not giving a pre-trial charge on the presumption of innocence.

Appellant specifically refers to only four panel members. He complains that two should have been stricken for cause but were not. But he concedes as to one of these that "she agreed that she could base her decision on the evidence and instructions by the Court." Thus she falls within the rule stated in *Brooks v. State*, 244 Ga. 574, 577 (261 SE2d 379) (1979) (death penalty vacated and case remanded for further consideration of sentence on other grounds, 446 U.S. 961 (1980)), quoting *Irvin v. Dowd*, 366 U.S. 717, 723 (81 SC 1639, 6 LE2d 751) (1961): "To hold that the mere existence of any preconceived notion as to the guilt or innocence of an accused, without more, is sufficient to rebut the presumption of a prospective juror's impartiality would be to establish an impossible standard. It is sufficient if the juror can lay aside his impression or opinion and render a verdict based on the evidence presented in court." This principle applies equally to a second juror of whom appellant complains.

Of the remaining two jurors, one was stricken for cause after being asked, "In other words, you think that the defendant has the burden to go forth, and if he gave you some evidence, he might change your mind?", and responding, "Right." The other did not answer the defendant's question, "Would you wait until the defendant gets to present his case before you fully made up your mind about the case?" because the state interposed an objection to the question. Appellant cites the instances involving these two jurors to show that the trial court impermissibly limited his voir dire; that is, had he been allowed to pursue such questions he could have shown that the prospective jurors were prejudiced against him. His argument, however, is not supported by the record. Upon the state's objection to the latter question being made, the trial court held a bench conference. Appellant's attorney did not respond to the state's argument that the question went to the burden of proof other than to say, "Surely I can ask the juror whether there's two sides to an issue?" The court responded affirmatively, and he then asked the juror: "In other words, what I was getting at, to be simple, you understand there's always two sides to an issue?" "Yes, sir." "And you

would hear both *806 sides before you make up your mind?" "Yes, sir." We find that appellant was permitted to ask the question objected to, albeit in different words.[2]

Appellant relies on *Decker v. State*, 139 Ga. App. 707, 708-09 (229 SE2d 520) (1976), to support his contention that the trial court erred in not giving a "pre-evidentiary charge" on such matters as the burden on the state to prove the defendant's guilt beyond a "reasonable doubt" and the defendant's presumption of innocence. *Decker*, however, merely holds that such a charge is a "salutary recent development." *Id.* at 709. It does not require that such a charge be given, and we likewise decline to do so. Appellant's argument that the failure of the trial court to give such a charge deprived him of a fair trial is without merit.

In a related enumeration of error, his twenty-eighth, the appellant contends that excessive prejudicial pre-trial publicity violated his right to a fair trial and to an impartial jury. Where prejudicial publicity exists to the extent that it will prevent a fair trial, the remedies are a continuance until the problem abates or a change of venue. *Sheppard v. Maxwell*, 384 U.S. 333, 363 (86 SC 1507, 16 LE2d 600) (1966). The appellant here sought neither. Rather he filed a motion seeking to have all charges dismissed, which the trial court properly denied.

The appellant did file a pre-trial motion entitled "Reserve Motion for Change of Venue." His attorney explained in a pre-trial conference that he was not seeking a change of venue at that time but rather filed the motion as a precautionary measure lest he determine during voir dire that a change of venue was necessary. He never pursued the motion, however, and immediately before the jurors were sworn the reserved motion was expressly withdrawn. We find no error here.

18). In appellant's thirtieth enumeration of error, he complains that the trial court did not allow him to file the results of a survey poll in support of his motions to dismiss and for change of venue due to prejudicial pre-trial publicity. While the survey was offered, unsolicited, to both the state and to the appellant, the results were not known to the parties because the pollster acceded to the trial court's request that this information not be released.

*807 As we said in Division 17, *supra*, regarding the pre-trial publicity, no motion for change of venue was pressed in this regard. Appellant now says in essence that he abandoned his motion because this survey poll, which would have proved his case, was denied him. In so arguing, he relies on *Wilson v. State*, 93 Ga. App. 229 (91 SE2d 201) (1956), wherein the only eyewitness to the case was in jail and the defense was denied access to the witness to interview him before trial. The Court of Appeals held that the trial

court's refusal to allow the defendant access, upon proper motion, constituted an abuse of discretion because it denied the defendant a fair trial and the benefit of counsel. The case before us is clearly distinguishable.

In *Wilson*, *supra*, the defendant was denied access to a material witness against him. Here, the appellant was able to conduct far ranging individual voir dire of all the veniremen regarding the effects of pre-trial publicity. "The single purpose for voir dire is the ascertainment of the impartiality of jurors, their ability to treat the cause on the merits with objectivity and freedom from bias and prior inclination." *Whitlock v. State*, 230 Ga. 700, 706 (198 SE2d 865) (1973). The appellant confronted each of the members of the jury personally. We find no error in the overruling of appellant's motion to release the public opinion survey.

19). In his thirty-second enumeration of error, appellant argues that the trial court erred in not considering four affidavits which he sought to admit at the hearing on his motion for new trial, all of which he argues were clearly sufficient to warrant the grant of a new trial on the basis of newly discovered evidence, and two of which he argues were necessary for the trial court to consider to be able to determine whether a violation of *Brady v. Maryland*, 373 U.S. 83 (83 SC 1194, 10 LE2d 215) (1963) had occurred at trial. We disagree.

To be granted a new trial on the basis of newly discovered evidence, the movant must "satisfy the court: (1) that the evidence has come to his knowledge since the trial; (2) that it was not owing to the want of due diligence that he did not acquire it sooner; (3) that it is so material that it would probably produce a different verdict; (4) that it is not cumulative only; (5) that the affidavit of the witness himself should be procured or its absence accounted for; and (6) that a new trial will not be granted if the only effect of the evidence will be to impeach the credit of a witness." *Timberlake v. State*, 246 Ga. 488 (1) (271 SE2d 792) (1980); *Drake v. State*, 248 Ga. 891 (1) (287 SE2d 180) (1982). All six of these requirements must be satisfied to secure a new trial. *Timberlake v. State*, *supra*, at 491. Furthermore, on appeal a trial court's denial of a motion for new trial will not be disturbed unless it affirmatively appears there was an abuse of discretion. *Drake v. State*, *supra*, at 894; *Findley v. State*, 251 Ga. 222 (4) (304 SE2d 898) (1983).

Appellant first argues that the trial court improperly failed to consider the affidavit of Stanley Parker, and that Parker's affidavit satisfied all six of the above requirements. Initially, we note that the trial court did admit and consider this affidavit, in which Parker states that he saw one of his friends stab Jo Jo Bell once in the back and twice in the chest. In the affidavit, he also claims that he and his friend then put a dark or light blue bedspread over the body, placed it in a pick-up truck, and drove it to a wooded area, where

they dumped it, poured gasoline on the bedspread, and "burned it up." We disagree with appellant's conclusion that this affidavit satisfies the six requirements enumerated in *Timberlake v. State*, *supra*. Specifically, even assuming that appellant could not have uncovered Parker's statement earlier, we find that this evidence is clearly not so material that it would probably produce a different verdict. *Timberlake v. State*, *supra*, at 496. First, the affidavit relates to a collateral issue: the uncharged crime of the murder of Joseph Bell. Second, Parker's credibility is weak because the affidavit's details concerning the murder of Bell differ vastly from the testimony presented at trial by the state concerning the cause of death and the postmortem treatment of the body, and because, a week after signing the affidavit, Parker told an Atlanta police officer that he had lied about being present when Bell was killed. These factors make it likely that Parker's testimony would not produce a different verdict, and, for this reason, the trial court did not err in denying appellant's motion for new trial to the extent it was based on Parker's affidavit.

Williams also complains of the trial court's failure to consider the affidavits of two experts whom he hired after his conviction to test samples from appellant's bedroom carpet. Williams alleges that their findings, as contained in their affidavits, see Division 7, *supra*, clearly demonstrate that the state withheld exculpatory fiber information from the defense. He concludes that the trial court erred in not considering these affidavits because, first, they constituted newly discovered evidence which was sufficient to warrant the grant of a new trial, and, second, their consideration was necessary for the trial court to be able to adequately resolve the merit of appellant's argument on motion for new trial that violations of *Brady v. Maryland*, *supra*, had occurred at trial. See Division 7, *supra*. Pretermittting the question whether the trial court should have considered these affidavits, see *Castell v. State*, 250 Ga. 776 (11) (301 SE2d 234) (1983), they were included in the material forwarded to this court, and, despite the trial court's refusal to consider them, we have carefully reviewed them, and find that they raise no meritorious *Brady* issue; hence, even if they had been admitted below, they would *809 not have led to the conclusion that violations of *Brady v. Maryland*, *supra*, had occurred at trial. Furthermore, these affidavits do not warrant the grant of a new trial because the findings therein could, with due diligence, have been discovered prior to trial. *Timberlake v. State*, *supra*, at 491.

Appellant further argues that the trial court erred in not considering the affidavit of Charles Morton, a defense expert who performed an independent analysis of the state's fiber evidence before, and during a part of, the trial. Williams argues that the affidavit, which he alleges shows the state substantially denied Morton access to the crime lab and to evidence kept there, was sufficient to warrant the grant of a new trial on the basis of newly discovered evidence. However, because Morton was appellant's own expert, and because

the appellant could have, in the exercise of due diligence, discovered this information during the course of the trial, this "newly discovered evidence" clearly does not satisfy the second requirement for the grant of a new trial. *Timberlake v. State*, 246 Ga. at 491.

For the above reasons, this enumeration of error is without merit.

20). In his thirty-third enumeration of error, Williams argues that he was denied effective assistance of counsel. However, appellant's trial counsel were not heard on this issue at the motion for new trial hearing, and for this reason, we decline to address the merits of this enumeration of error. *Simpson v. State*, 250 Ga. 365 (2) (297 SE2d 288) (1982); *Brown v. State*, 251 Ga. 598 (308 SE2d 182) (1983).

21). We have examined appellant's remaining enumerations of error and find them to be without merit. *Kilgore v. State*, 251 Ga. 291 (8) (305 SE2d 82) (1983); *Myron v. State*, 248 Ga. 120 (10) (281 SE2d 600) (1981).

Judgment affirmed. All the Justices concur, except Smith, J., who dissents.

SMITH, Justice, dissenting.

I respectfully dissent. I am convinced that appellant did not receive a fair trial, primarily because of the admission into evidence *810 of ten prior crimes for which he was never charged. The ten extrinsic offenses were highly prejudicial, did not fit any recognized exception to the general rule excluding such evidence, and were never clearly proven by the state. I would reverse and remand for a new trial.

1. The majority opinion affirms the trial court's admission into evidence of 10 killings for which Williams was never charged, based on the "victims' commonality of appearance and behavior," Williams' alleged "negative attitude" toward poor young blacks, and the majority's conclusion that "the sheer number of victims with common characteristics each logically connected with Williams by hairs and fibers tends to show a pattern of killings." I disagree. The trial court's admission of the highly prejudicial other crimes evidence was erroneous under any fair reading of our prior decisions, and the majority's approval of the other crimes evidence in this case distorts beyond recognition any exception to the general rule excluding evidence of other crimes committed by the accused.

"On a prosecution for a particular crime, evidence which in any manner shows or tends to show that the accused has committed another crime wholly distinct, independent, and separate from that for which he is on trial, even though it be a crime of the same sort, is irrelevant and inadmissible, unless there be shown some logical connection between the

two from which it can be said that proof of the one tends to establish the other." *Bacon v. State*, 209 Ga. 261 (71 SE2d 615) (1952). The reasons for excluding evidence of other crimes are well known. "Courts that follow the common-law tradition almost unanimously have come to disallow resort by the prosecution to any kind of evidence of a defendant's evil character to establish a probability of his guilt ... The state may not show defendant's prior trouble with the law, specific criminal acts, or ill name among his neighbors, even though such facts might logically be persuasive that he is by propensity a probable perpetrator of the crime. The inquiry is not rejected because character is irrelevant; on the contrary, it is said to weigh too much with the jury and to so overpersuade them as to prejudge one with a bad general record and deny him a fair opportunity to defend against a particular charge. The overriding policy of excluding such evidence, despite its admitted probative value, is the practical experience that its disallowance tends to prevent confusion of issues, unfair surprise and undue prejudice." *Michelson v. United States*, 335 U.S. 469, 475-76 (69 SC 213, 93 LE 168) (1948). An exception to the general rule excluding other crimes evidence arises where such evidence "is substantially relevant for some other purpose than to show a probability that [the defendant] committed the crime on trial because he is a man of criminal character," *Walraven v. State*, 250 Ga. 401, 407 (297 SE2d 278) *811 (1982).

Purposes for which other crimes evidence may be offered include motive, intent, absence of mistake or accident (both are aspects of intent), plan or scheme, and identity. *Id.* at 408. Our cases state clearly that for evidence of extrinsic offenses to be admissible for one or more of these purposes the state must establish that: (1) the defendant was the perpetrator of the extrinsic offense, and (2) there is a sufficient similarity or connection between the extrinsic offense and the offense charged, such that proof of the former tends to prove the latter. *Walraven v. State*, *supra*; *Kilgore v. State*, 251 Ga. 291, 296 (305 SE2d 82) (1983); *Hamilton v. State*, 239 Ga. 72, 73 (235 SE2d 515) (1977). Moreover, (3) "[i]t has long been the rule in Georgia that evidence of an independent crime is never admissible unless the prejudice it creates is outweighed by its relevancy to the issues on trial." *Robinson v. State*, 246 Ga. 469, 470 (271 SE2d 786) (1980). See *Tuzman v. State*, 145 Ga. App. 761, 763 (244 SE2d 882) (1978); *Carroll v. State*, 143 Ga. App. 796 (240 SE2d 197) (1977).

The second prong of this test is in reality a specific application of the general requirement that evidence be logically relevant to a disputed issue at trial. *Walraven v. State*, *supra*, at 407. Depending on the issue to which the extrinsic offense is addressed, the state may be required to prove a high degree of similarity between relevant characteristics of the extrinsic offenses and the charged crimes, or it may only have the burden of showing a logical connection between crimes which are essentially dissimilar. *Bacon v. State*, *supra*, at 263; *State v. Johnson*, 246 Ga. 654, 657 (272 SE2d 321) (1980) (Hill, J., dissenting); *United*

States v. Beechum, 555 F2d 487 (5th Cir. 1977), vacated on other grounds, 582 F2d 898 (1978).

The majority opinion, while undertaking an exhaustive factual review of the details of other crimes evidence introduced against Williams, wholly fails to identify the purpose for which the other crimes evidence was introduced, or to apply to its fact statement the well-established three-part test for admissibility. Instead, the opinion hints vaguely that proof of some sort of "pattern" of mass murders of Atlanta black youth emerges from the evidence and that this pattern justified admission of the other crimes evidence.[1] Under *812 its shadowy "pattern" rubric, the majority opinion approves of the wholesale admission, in a highly publicized and emotionally charged trial, of ten uncharged homicides which, admittedly, bear varying degrees of similarity to the Payne and Cater killings for which Williams was on trial. The majority does so with very little explanation of its basis for decision, managing to do violence to our prior rule governing admission of other crimes evidence, and to confuse both bench and bar at the same time.

(a) In this case, the only arguable purposes for introducing evidence of the ten uncharged offenses were to show Williams' identity as the killer by proving "plan or scheme" and "modus operandi."

To be admissible under the "plan or scheme" exception, the extrinsic offense and the crime charged must be connected in the sense that they have "such a concurrence of common features that the various acts are naturally to be explained as caused by a general plan of which they are the individual manifestations." *United States v. Goodwin*, 492 F2d 1141, 1153 (5th Cir. 1974) (quoting Wigmore). In other words, the crimes must be mutually dependent, so that one naturally flows from and can be explained by another. See *Natson v. State*, 242 Ga. 618, 620 (250 SE2d 420) (1978); *United States v. O'Connor*, 580 F2d 38, 41 (2d Cir. 1978). In this context evidence of other crimes must be offered to show the accused's identity as the perpetrator of the charged crimes by showing that he acted in accordance with a larger goal, not merely that he acted with a propensity to commit crimes generally. See *Wright & Graham, Federal Practice and Procedure* 5244, pp. 500-501 (murder of three fellow heirs to estate by defendant, each by a different method, admissible to show "plan or scheme" in trial for murder of third co-heir).

In this case there was obviously no such connection between the two charged crimes and the ten extrinsic offenses sufficient to justify their admission to prove a larger plan or goal of which the murders of Payne and Cater were a part. Even assuming that Williams was proven to be the perpetrator of the ten extrinsic offenses (an assumption which I am unwilling to make, see pp. 816-17, *infra*), the ten extrinsic offenses show, at best, a "plan"

with no definable goal *813 except a desire to randomly kill mostly young, poor black males. In such a "plan," all the crimes were independent and separate, not mutually dependent, they occurred at different times over a 2-year period, and no single crime was necessary to bring any goal to fruition. The deaths of Payne and Cater are not logically explained as a result of any larger "plan" of Williams, and the only inference that can be drawn in this regard from the admission of the other crimes evidence is the forbidden one; namely that, because Williams had a propensity to kill young, poor black males, he acted consistently with this propensity in killing Payne and Cater. See *Walraven v. State*, supra, at 407; *Wright & Graham*, supra, at 500. Because "plan or scheme" is simply inapplicable here, I shall not further discuss whether the state's evidence on this point met the three-part test for admissibility of other crimes evidence.

(b) The only remaining legitimate purpose for which the state could have offered the other crimes evidence was to show that Williams employed a singularly unique *modus operandi*[2] when committing all twelve murders. Examination of the record indicates that proving Williams' *modus operandi* (and hence his identity) was in fact the state's main purpose in introducing the other crimes evidence at trial. This issue, then, should be carefully analyzed in light of the state's burden of showing similarity or logical connection, perpetration, and legal relevance.

(1) As to similarity, it is a longstanding premise of evidence law that in order to prove the accused's identity by use of *modus operandi* evidence, the extrinsic offense and the charged crime must bear an extremely high degree of similarity. See *Sweeny v. State*, 152 Ga. App. 765, 767 (264 SE2d 260) (1979) (extrinsic offense and charged crime "sufficiently bizarre and analogous" to allow admission of other crimes evidence); *United States v. Beechum*, 582 F2d 898, 912 n. 15 (5th Cir. 1978); *United States v. Myers*, supra, at 1045; *Wright & Graham*, supra, 5240, p. 513. McCormick states that the other crimes must be "so nearly identical in method as to earmark them as the handiwork of the accused. Here much more is demanded than the mere repeated commission of crimes of the same class, such as repeated burglaries or thefts. The device used must be so unusual and distinctive as to be like a signature." McCormick on Evidence, 190, p. 449.

Examining the majority opinion's factual review of the Payne *814 and Cater murders and the ten uncharged offenses, one is indeed struck by several similarities among the twelve crimes. Each of the victims was a low-income black male, slightly built, who was often seen alone in the streets of Atlanta. Payne, Cater, and five of the ten other crimes victims were seen with Williams sometime prior to their death. All but two of the victims, Porter and Middlebrooks, were killed by some form of asphyxiation.

Just as important as these similarities, in my opinion, however, are substantial dissimilarities between the two charged offenses and the ten extrinsic crimes. Payne and Cater, age 21 and 28, respectively, were adults; the ages of the victims of the uncharged crimes ranged from 11 years to 28 years and averaged only 15.7 years. With the exception of 28-year-old John Porter, the extrinsic offense victims were essentially children. Another striking dissimilarity between the Payne and Cater killings, on the one hand, and the ten extrinsic offenses, on the other, is that while the bodies of Payne and Cater were both apparently thrown into the Chattahoochee River near the I-285 overpass, only one of the ten extrinsic offense victims' bodies, that of Joseph Bell, was found in a river. Bell's body was discovered in the South River near Rockdale County, miles from where Payne and Cater were found. The remaining nine were deposited on land. Although there was evidence tending to show that the Cater killing was sexually motivated, there was absolutely no medical evidence showing sexual abuse of any of the victims. In addition, it is critical to note that the state's fiber evidence allegedly linking Williams to all twelve victims, while slightly probative on the issue of whether Williams actually perpetrated the ten other crimes, see pp. 816-17, *infra*, has no relevance to the *modus operandi* issue, for the simple reason that the fiber evidence in this case provides no information as to the murderer's technique in killing or disposing of his victims. The state's own experts testified that they could not determine the exact mechanism of the alleged transfer of fibers from Williams to the victims. Thus, the sole implication of this type of trace evidence is that each of the victims possibly was in contact with Williams, his house, or his car sometime before his death. Although this inference may be probative of the identity of the killer of the ten extrinsic victims, it does not establish a unique *modus operandi*, since it would be possible for the murderer to apprehend, kill, and dispose of his twelve victims in dissimilar ways, yet transfer fibers to them in each case. See *United States v. Myers*, 550 F2d 1036, 1045 (5th Cir. 1977). Thus the presence or absence of fiber evidence has no relevance in the case before us to the narrow issue of *modus operandi*. Finally, the evidence that Wayne Williams had been sighted at the funerals of Pue, Geter and Rogers, while an interesting sidelight, has no *815 relevance here, since this is not a feature shared by either the Payne or Cater cases.

Comparing the relative similarities and differences of the ten extrinsic offenses, it is evident to me that only one — the Joseph Bell killing — is "so nearly identical in method" to the charged crimes as to possibly be admissible as evidence of Williams' *modus operandi*. Like Payne and Cater, Bell was seen with Williams shortly before his disappearance. He was killed by asphyxiation, a relatively rare form of homicide. Although dissimilar to Payne and Cater in the sense that he was only 15 years old when killed, he was, like them, a black male with a "street kid" lifestyle whose body was dumped in a river by the killer. These

circumstances make the Bell killing substantially unique, yet similar enough to the Payne and Cater homicides, so as to be probative of *modus operandi*.

The Pue, Geter, Rogers and Barrett crimes, on the other hand, are of questionable relevance and should have been excluded by the trial judge. While sharing many characteristics of the charged offenses (particularly asphyxiation), all four of these victims were dumped by the killer on land, not in water. In my view this fact distinguishes them from the Payne and Cater murders, making them less relevant on the *modus operandi* issue.

A third group of victims, consisting of Evans, Stephens, and Baltazar, are even less relevant to the issue of *modus operandi* than the Pue, Geter, Rogers and Barrett crimes. Although each was a young black male who was killed by asphyxiation, substantial dissimilarities exist between these cases and the Payne and Cater murders. The bodies of Evans and Stephens were discovered partially clothed, and Baltazar was found fully clothed. Each was deposited on land, not in water. In addition, neither Evans, Stephens, nor Baltazar was ever seen with Wayne Williams. Weighing these factors, while considering the highly prejudicial impact of such testimony, I conclude that evidence of the Evans, Stephens, and Baltazar killings should have been kept from the jury.

As to the Middlebrooks and Porter extrinsic offenses, it is clear that neither bore the requisite degree of similarity to the charged crimes. Both Middlebrooks and Porter were found on land, fully clothed, but, more significantly, neither was ever seen with Williams and neither was killed by asphyxiation. Accordingly, neither is even slightly probative of *modus operandi*, and their admission into evidence was error.

Therefore, I would reverse the conviction on the ground that nine of the ten extrinsic offenses were not sufficiently similar or logically connected to the two charged offenses and were thus inadmissible to show *modus operandi*. Although this ground is *816 sufficient for reversal, for purposes of clarity I will also address the two remaining prongs of the test for relevancy of other crimes evidence ÔÇö whether Williams was the perpetrator of the extrinsic offenses, and whether the probative value of the other crimes evidence outweighed its prejudicial impact.

(2) The evidence tending to show that Williams perpetrated the ten extrinsic offenses was purely circumstantial and can only be characterized as weak. Apart from the sightings of five of the ten extrinsic offense victims with Williams prior to their deaths, the only evidence linking Williams to any of the ten other crimes consisted of two blood stains and the state's dog hair and fiber evidence, which was of questionable reliability and probative

worth (see Division 2, *infra*). The majority opinion admits that the evidence showing that Williams perpetrated the extrinsic crimes "was stronger in some cases than others," but maintains that "in each case there was some evidence connecting Williams with the victim." (Emphasis supplied.) I submit that a mere showing of "some evidence" that the accused committed the other crimes sought to be used against him has never been sufficient (at least until today's decision) for admission of evidence of extrinsic offenses.

While this court has never squarely addressed the issue of the proper standard of proof for "other crimes" evidence,[3] I would require introduction of "clear and convincing" evidence that the accused committed the extrinsic offenses. See *McCormick*, *supra*, 190, p. 452; *Wright & Graham*, *supra*, 5249, p. 532; *Sweeny v. State*, *supra*, ("clear proof"). This appears to be the predominant view in other jurisdictions. See *State v. Botham*, 629 P2d 589 (Colo. 1981); *Page v. United States*, 438 A2d 195 (D. C. App. 1981); *Chapman v. State*, 417 S2d 1028 (Fla. App. 1982); *State v. Lee*, 381 S2d 792 (La. 1980); *Cross v. State*, 386 A2d 757 (Md. App. 1978); *State v. McAdoo*, 330 NW2d 104 (Minn. 1983); *State v. Wilson*, 385 A2d 304 (N. J. Super. 1978); *State v. Conyers*, 233 SE2d 95 (S. C. 1977); *Sanders v. State*, 604 SW2d 108 (Tex. Ct. App. 1980); *United States v. O'Brien*, 618 F2d 1234 (7th Cir. 1980); *United States v. Engleman*, 648 F2d 473 (8th Cir. 1981); *United States v. Bailleaux*, 685 F2d 1105 (9th Cir. 1982). A relatively high standard of proof is necessary because of the questionable relevancy of other crimes evidence and its devastating *817 effect on a defendant's case.[4] Such a standard is particularly appropriate where, as here, the alleged perpetrator was never charged with or convicted of the other crimes, and the evidence of his commission of them is purely circumstantial. Introduction of a prior conviction would, of course, satisfy this standard.

The fragmentary and wholly circumstantial evidence introduced by the state on this point can hardly be characterized as "clear and convincing." I must take issue with the majority's approach to this issue, which seems to be that, although the evidence showing that Williams was the perpetrator of several of the extrinsic offenses was quite weak, the ten crimes as a group were admissible because of their "sheer number," and because the existence of a perceived "pattern" of killings obviates the necessity of proving that Williams was, in fact, the perpetrator of each extrinsic offense. See, e.g., *United States v. Woods*, 484 F2d 127 (4th Cir. 1973). This "cumulation" theory of admitting other crimes evidence is at odds with the rule of evidence which requires clear proof of the accused's commission of each and every extrinsic offense. *Wright and Graham* note that "[s]ome cases seem to admit evidence of other crimes on a slender showing of the defendant's connection with the crime where the crimes are numerous, apparently on some sort of cumulation theory. This approach should be used sparingly, as it seems capable of producing unjust results;

defendant's suspicious connection with a number of crimes does not necessarily make him guilty of any of them." Wright & Graham, *supra*, 5246, p. 514.

As to the Porter and Middlebrooks extrinsic offenses, the evidence tending to show Williams was the perpetrator was weak, consisting of a very few matching fibers and dog hairs, and a blood stain consistent with Porter's blood type found in Williams' car. The fact that these two crimes could be admitted on such scant evidence illustrates the basic unfairness of this trial and Williams' unenviable position as a defendant who, charged with two murders, was forced to defend himself as to twelve separate killings. The state's case on other crimes evidence was built on innuendo, suspicion and guilt by association, not on clear and convincing proof, and the admission of the other crimes evidence in this case was tantamount to a "bootstrap operation" which allowed the state to bypass the requirement that it prove each of the twelve offenses sought to be used against appellant. *818 See *United States v. Woods*, 484 F2d 127, *supra* (Widener, J., dissenting).

(3) The final prong of the three-part standard for admissibility of other crimes evidence is the balancing of the proffered evidence's probative value against its potential for prejudice to the defendant. The majority opinion completely sidesteps this issue, which is considered crucial by most courts and commentators. See *Robinson v. State*, 246 Ga. 469, 470 (271 SE2d 786) (1980); *Tuzman v. State*, *supra*; *Carroll v. State*, *supra*; *United States v. Beechum*, *supra*, 582 F2d at 913-14; *McHale v. United States*, 398 F2d 757 (D. C. Cir. 1968); Wright & Graham, *supra*, 5249, pp. 533-38, and 5250, pp. 541-551; McCormick, *supra*, 190, pp. 452-454; Note, Other Crimes Evidence at Trial: Of Balancing and Other Matters, 70 Yale L. J. 763 (1961).

In *Carroll v. State*, *supra*, the Court of Appeals stated: "The prejudicial effect of evidence concerning independent crimes is the paramount consideration behind the general rule of inadmissibility of such evidence ... Even [if independently relevant to an issue in the case], the evidence will not be admissible unless its relevance to the issue outweighs its prejudicial impact. [Cit.] If the evidence tends to show a general criminal propensity more than it tends to prove an issue in the case, it should not be introduced to the jury." *Id.* at 797. McCormick's advice bears repeating here: "[T]he problem is not merely one of pigeonholing, but one of balancing, on the one side, the actual need for the other-crimes evidence in the light of the issues and the other evidence available to the prosecution, the convincingness of the evidence that the other crimes were committed and that the accused was the actor, and the strength or weakness of the other-crimes evidence in supporting the issue, and on the other, the degree to which the jury will probably be roused by the evidence to overmastering hostility." McCormick, *supra*, 190, p. 453. In this regard, the

trial court must weigh all these elements and exercise its discretion in deciding whether to admit or exclude the proffered other crimes evidence. "It should be recognized, however, that this is not a discretion to depart from the principle that evidence of other crimes, having no substantial relevancy except to ground the inference that [the] accused is a bad man and hence probably committed this crime, must be excluded. The leeway of discretion lies rather in the opposite direction, empowering the judge to exclude the other-crimes evidence, even when it has substantial independent relevancy, if in his judgment its probative value for this purpose is outweighed by the danger that it will stir such passion in the jury as to sweep them beyond a rational consideration of guilt or innocence of the crime on trial. Discretion implies not only leeway but *819 responsibility." *Id.* at 453-54.

The record does not show that the trial judge responsibly exercised his discretion in this case either by assessing the relative probative strength of the ten extrinsic offenses, or by determining whether the other crimes evidence was reasonably necessary to the state's case. See *United States v. Kirk*, 528 F2d 1057, 1060 (5th Cir. 1976). Instead he allowed into evidence all ten extrinsic crimes sought to be introduced by the state, regardless of the factual dissimilarities between the ten crimes, the state's sketchy proof that the defendant committed each crime, and the extremely prejudicial impact of admitting evidence of the extrinsic offenses. The majority opinion concludes that the "sheer number" of other crimes, all with some degree of similarity to the Payne and Cater killings, makes them admissible as evidence of a "pattern" of murders. But the "sheer number" of other crimes introduced in this case and their prejudicial weight are the very reasons the evidence should have been excluded. Each additional crime had an incremental prejudicial effect and contributed to confusion of the issues and undue consumption of trial time, with very little countervailing probative worth. See *United States v. Gaus*, 471 F2d 495, 499 (7th Cir. 1973) (proof of eleven similar offenses held inadmissible because not necessary to the state's case); *People v. Schlatter*, 390 NYS2d 441 (1977) (proof of 40 similar offenses inadmissible because of extreme prejudicial effect). In this case, several factors, including the questionable probative worth of the fiber and dog hair evidence; the availability of other proof as to the Payne and Cater killings; the weakness of the proof of other crimes, especially as to Porter and Middlebrooks; the heinous nature of the other crimes proved; and the inordinate amount of trial time (approximately one half) devoted to proof of the extrinsic offenses, all militate toward the conclusion that the prejudicial impact of the other crimes evidence outweighed its probative worth. See *United States v. O'Connor*, *supra*, at 43 (other crimes evidence "distorted the emphasis at trial away from the crimes covered by the indictment to those not so charged.")

(c) A wholly independent ground on which the other crimes evidence should have been excluded is that its introduction, in the fourth full week of Williams' trial, unfairly surprised appellant and deprived him of an opportunity to adequately prepare his defense as to the ten extrinsic offenses. The record shows that Williams contemplated that evidence of some 27 extrinsic offenses might be used against him at trial and that he asked for and received Brady materials on each of the 27 other crimes prior to trial. The trial began December 28, 1982. It was not until January 22, 1982, midway through the trial, that Williams was informed which of the 27 *820 extrinsic offenses would be introduced into evidence against him. During this entire period, the state was conducting scientific tests which were used against Williams at trial. Thus, from the time of his arrest on June 21, 1981 until January 22, 1982, Williams was for all practical purposes defending himself against 29 murder charges. Such a procedure flies in the face of any concept of an orderly, fair trial process in which a defendant is put on notice of the charges against him and allowed time to adequately prepare a defense. This court should, at a minimum, require the state to give pre-trial notice of its intention to introduce other crimes evidence against a defendant. Other state courts have recognized the need for such a pre-trial notice requirement. See *State v. Acquin*, 381 A2d 239 (Conn. Super. 1977); *State v. Germain*, 433 S2d 110 (La. 1983); *State v. Just*, 602 P2d 957 (Mont. 1979); *Burks v. State*, 594 P2d 771 (Okla. Cr. App. 1979); *Gipson v. State*, 619 SW2d 169 (Tex. Cr. App. 1981); *State v. Nicholson*, 252 SE2d 894 (W. Va. 1979). An acceptable procedure was outlined in *Burks v. State*, supra: (1) 10 days prior to trial, the state must furnish the defendant with a written statement of the extrinsic offenses it intends to prove at trial; (2) at the time the evidence is offered, the state must specify the exception under which the evidence is sought to be admitted; (3) there must exist a strong similarity or connection between the charged offenses and the other crimes; (4) the state must make a showing of necessity and that the evidence is not merely cumulative; (5) the evidence that the defendant committed the other crimes must be clear and convincing; and (6) limiting instructions must be given. In my view a pre-trial notice procedure is essential to the fairness of trials in which the state introduces evidence of other crimes.

The rule excluding other crimes evidence is not a mere technical rule of evidence, violation of which can be dismissed by incantation of the phrases "pattern" or "harmless error." Rather, the rule implicates basic constitutional rights to due process and a fundamentally fair trial. See *Lovely v. United States*, 169 F2d 386, 389 (4th Cir. 1948); *Bray*, Evidence of Prior Uncharged Offenses and the Growth of Constitutional Restrictions, 28 U. Miami L. Rev. 489 (1974). As the Supreme Court stated in *Brinegar v. United States*, 338 U.S. 160, 174 (69 SC 1302, 93 LE 1879) (1949): "Guilt in a criminal case must be proved beyond a reasonable doubt and by evidence confined to that which long experience in the common-

law tradition, to some extent embodied in the Constitution, has crystallized into rules of evidence consistent with that standard. These rules are historically grounded rights of our system, developed to safeguard men from dubious and unjust convictions, with resulting forfeitures of life, liberty and property." These safeguards were denied here.

*821 2. I cannot join Division 1 of the majority opinion, which purports to deal with the issue of the scientific reliability of the fiber evidence introduced at trial. Despite the novelty of this issue and the central role this type of evidence played in the state's case, a majority of the court today approves of the admission of this evidence almost without discussion, and apparently holds that fiber evidence has reached a stage of scientifically verifiable certainty.[5] I do not agree. At trial, various state experts testified concerning their recovery and analysis of certain fibers discovered on the bodies of the twelve victims and in the Williams home and car. Based on their analyses, they opined that some of the fibers found on the victims displayed characteristics which were consistent with the characteristics of fibers from the Williams environment, and further inferred that the fibers associated with the victims could have been transferred from the Williams home and car. The experts described criteria they used to assess the significance of the matches resulting from the fiber comparisons and, applying these criteria, concluded that it was "virtually impossible" for the fibers obtained from the bodies of eleven of the twelve victims[6] to have originated anywhere other than the Williams environment. My review of the record, however, indicates that the state failed to lay a foundation sufficient to establish that the methodologies its experts used to draw their inferences of significance are scientifically valid. For that reason I would hold that the trial court erred when it allowed the introduction of the fiber evidence.

In *Harper v. State*, 249 Ga. 519, 525 (292 SE2d 389) (1982), we stated that "it is proper for the trial judge to decide whether the procedure or technique in question has reached a scientific stage of verifiable certainty, or ... `rests upon the laws of nature.' The trial court may make this determination from evidence presented to it at trial by the parties; in this regard expert testimony may be of value. Or the trial court may base its determination on exhibits, treatises or the rationale of cases in other jurisdictions. [Cits.] The significant point is that the trial court makes this determination based on the evidence available to him rather than by simply calculating the *822 consensus in the scientific community. Once a procedure has been recognized in a substantial number of courts, a trial judge may judicially notice, without receiving evidence, that the procedure has been established with verifiable certainty, or that it rests upon the laws of nature."

In this case it does not appear that any decisions of other jurisdictions were brought to the trial judge's attention during the trial or that the judge ever made a specific finding that the techniques of assessing the significance of fiber comparisons used by the state's experts have reached a "scientific stage of verifiable certainty," *Harper v. State*, supra at 525. On appeal the state has cited cases to support its contention that fiber comparisons are competent evidence, but these cases are not on point. Although some of them support the proposition that present techniques and theories of fiber recovery and comparison are sufficiently verifiable to permit the drawing of an opinion that a certain fiber may have originated from a particular source, none addresses the problem of the verifiability of the methods used by experts to venture beyond those assessments of significance and state that there is a high degree of certainty that a particular fiber did in fact come from a particular source. Furthermore, my research reveals no cases which specifically address this problem. For this reason it was not possible for the trial court (or for this court) to judicially notice the verifiable certainty of the procedures used to estimate the significance of fiber matches testified to by experts at trial. Instead, the competency of the expert estimates of significance in this case must be proven, if at all, by the evidence received at trial.

The record shows that the key to the state's fiber evidence case was expert testimony concerning the alleged uniqueness of two types of carpet fibers recovered and analyzed by the state's experts. These "unique" fibers were identified by the state's experts as (1) the green nylon carpet in appellant's bedroom, and (2) the green-black rayon floorboard carpet of the 1970 Chevrolet station wagon Williams was driving on May 22, 1981, the night of the Jackson Parkway bridge incident.

As to the bedroom carpet, there was testimony to establish that it was composed of fibers which were produced by a Boston, Massachusetts, company named Wellman, and that this type of fiber was sold to a Dalton carpet manufacturer named West Point Pepperell, and to as many as six other Georgia carpet manufacturers during the period in question. A Wellman employee testified that the fiber, known as 181-b, had an unusual shape (trilobal, with two long lobes and one short lobe), which was designed to avoid infringing upon a patented DuPont equilateral trilobal shape. *823 This employee also identified an exhibit which he said was a summary of Wellman's sales of the 181-b during the late 1960's to early 1970's.

A West Point Pepperell employee named Baggett testified that most of West Point's manufacturing of carpets using the 181-b had taken place in 1971, but that because the company would have continued to use any remaining inventory of that fiber until it was exhausted, production with that fiber would have continued for some undetermined time

subsequent to 1971, and sales of carpets fabricated with the fiber could have occurred as late as 1973. The West Point employee said that his company had manufactured several carpet lines with the 181-b but that he had experienced considerable difficulty in ascertaining sales figures for those lines because records of sales were usually discarded after seven years. However, although total sales for all lines using the 181-b could not be established, he had located the records for several lines, including two known as Luxaire and Dreamer.

Baggett said that Luxaire was produced in several colors, including one using a dye formulation named English Olive, and that Dreamer was also colored with this dye. He identified two pieces of carpet as samples from the company's records of Luxaire dyed to the English Olive hue, and also identified a copy of his company's sales records for Luxaire and Dreamer English Olive in the years 1971 and 1972. Baggett also gave his opinion whether another exhibit identified as a piece of the green carpet from the Williams bedroom was similar to the samples of Luxaire English Olive from his company's records. Although he admitted he was not qualified to perform microscopic analysis of single fibers, he testified that based upon his visual inspection of aggregate physical characteristics such as height of pile, weight of carpet, and type of backing, the carpets appeared to be similar.

Harold Deadman, an FBI microanalyst, identified another piece of carpet as a sample the FBI had obtained from West Point Pepperell. Deadman testified that West Point Pepperell had informed the FBI that the carpet sample was a piece of the Luxaire English Olive. He compared the gross physical characteristics of this exhibit with another piece of the Williams bedroom carpet and concluded that there were no significant differences in their construction and that they were probably manufactured by the same company. Then, building on the Luxaire and Dreamer sales records of West Point Pepperell, information orally supplied him by Baggett, and housing statistics provided by the Atlanta Regional Commission, as well as a number of wholly speculative assumptions (chief of which was that the Williams carpet was in fact a West Point Luxaire or *824 Dreamer English Olive carpet),[7] Deadman attempted to use the calculus of compound probabilities to perform a series of calculations to establish the rarity of that type of carpet in the Atlanta metropolitan area. See McCormick, *supra*, 204. He finally concluded that there was a one in 7792 chance of randomly selecting a home in the Atlanta area and finding a room containing carpet similar to the Williams bedroom carpet.

Regarding the green-black 1970 Chevrolet carpet, both Deadman and Peterson testified that they had information indicating that in the Atlanta area only 620 out of over two

million cars had that type of carpet. Deadman explained that this data had been supplied by General Motors.

During arguments to the jury an assistant district attorney summarized the foregoing testimony and embellished his summary with his own attempt to quantify the probative force of the fiber evidence. Accordingly, he rounded off the figures for the 181-b bedroom carpet and the green-black floorboard carpet and multiplied them together in order to calculate the chances "that there is another house in Atlanta that has the same kind of carpet as the Williams house and that the people who live in that house have the same type station wagon as the Williamses do ...," arriving at a probability of one in forty million. Adjusting this figure to account for an additional assumption of his own, the prosecutor argued that the appropriate figure was actually one in one hundred fifty million.

Taken at face value, the testimony establishing the rarity of the two fiber types might seem to be a sufficient underpinning for the ultimate conclusion of the experts that the fibers of those types found on the bodies were probably transferred from the Williams home or car. Closer examination reveals, however, that the testimony suffers from fatal evidentiary and constitutional defects, and is in fact worthless as evidence on the issue of scientific admissibility of the expert's opinions of significance.

Much of the state's failure of proof in this respect resulted from its experts' insistence on incorporating hearsay into their testimony setting out the factual bases for their opinions. Although an expert may base his opinion on facts he himself has observed or on facts testified to by other witnesses, in Georgia the rule is that hearsay, even if admitted without objection, has no probative value and cannot be considered to sustain a verdict. Green, Ga. Law of Evidence *825 (2d ed. 1983) 111 & 222; Agnor's Ga. Evidence (1976) 9-7 & 11-3. In recent criminal cases, we have reaffirmed the principles that it is improper for an expert witness to repeat the hearsay opinion of another expert, Dean v. State, 250 Ga. 77 (295 SE2d 306) (1982), and that an expert may not base his opinion upon hearsay facts which have not been admitted into evidence, Vaughn v. State, 249 Ga. 803, 805 (294 SE2d 504) (1982). Only one of our decisions appears to run contrary to this rule. In King v. Browning, 246 Ga. 46 (268 SE2d 653) (1980), a land line case, we stated that "[a]n expert may base his opinion on hearsay and may be allowed to testify as to the basis for his findings ..." Id. at 47. Whatever the impact of the ruling in King v. Browning on civil cases, it is clear that in criminal cases allowing an expert to testify based solely on hearsay violates the constitutional right of a criminal defendant to confront the witnesses against him. See Stewart v. State, 246 Ga. 70 (4) (c) (268 SE2d 906) (1980); Harrell v. State, 241 Ga. 181

(243 SE2d 890) (1978); *Ohio v. Roberts*, 448 U.S. 56 (100 SC 2531, 65 LE2d 597) (1980); Agnor, *supra*, at 11-4; Green, *supra*, at 224.

Applying the well-established rule which disallows the probative value of hearsay evidence, it becomes clear that the state's attempts to prove uniqueness of the Wellman fibers and the automobile carpeting were totally inadequate. First, the sales figures from Wellman and West Point Pepperell are obvious hearsay which were not brought within the business records exception to the hearsay rule, OCGA 24-3-14 (Code Ann. 38-711); see Green, *supra*, Ch. XX, and thus there is no competent evidence about the total amount of Wellman 181-b fiber or West Point Pepperell Luxaire and Dreamer English Olive carpet produced. Second, the two Luxaire English Olive carpet samples, as well as the sample previously supplied to the FBI and identified by Deadman, were not admissible into evidence because each one was in fact a business record which incorporated not just the physical piece of carpet, but also the attendant written corporate record identifying it, with none of the three being qualified as a business record exception to the hearsay rule. Accordingly, there were no samples of the Luxaire properly in evidence with which to compare the Williams carpet, and no probative inferences could be drawn identifying the Williams carpet as Luxaire English Olive, or for that matter as a West Point Pepperell product. These gaps in the state's proof, together with the hearsay nature of the statistics from the Atlanta Regional Commission, made a sham of the state's theory of fiber uniqueness. Deadman's mathematical calculations were particularly worthless, in light of the fact that they were in several instances based not only upon hearsay evidence but also upon his own unproven assumptions. The weakness *826 of the Wellman fiber evidentiary prop, coupled with the hearsay character of the General Motors data as to the commonness of the green-black floorboard carpet, destroys any basis for multiplying probabilities in the manner of the prosecutor's closing argument to arrive at a fictional one-in-millions probability that any other Atlanta household had the same type of bedroom and auto floorboard carpets. This evidence, which was extremely prejudicial to Williams, should never have been admitted.

Thus the only competent evidence pertaining to fibers was the proof of the recovery and comparison of fibers, and the testimony that fibers found on the victims appeared similar to fibers found in the Williams home and car and could have had a common origin. The remaining facts and inferences were rank hearsay, unproven assumptions, and guesswork, and should not have been admitted by the trial court. Since the evidence of guilt in this case was solely circumstantial and because the ultimate conclusion of significance of the fiber matches was repeatedly placed before the jury in terms expressing a very high degree of confidence, the error in admitting the fiber evidence and testimony in all probability contributed to the jury's verdict.

3. I also disagree with the conclusion reached in Division 3 of the majority opinion involving appellant's request for scientific reports. The majority relies on *Law v. State*, 251 Ga. 525 (307 SE2d 904) (1983), for the proposition that only written scientific reports must be given over to a defendant who makes a timely request pursuant to OCGA 17-7-211 (Code Ann. 27-1303). However, the dissents of Justice Weltner and myself in *Law* point out the precise problem presented here. Whereas in *Law* the state contended that exigent circumstances prevented introduction of written scientific report evidence due to time limitations, the present case is clearly an example of the kind of manipulation warned against in the *Law* dissents.

On August 14, 1981, appellant made a proper and timely request under OCGA 17-7-211 (a) (Code Ann. 27-1303) for production of scientific reports. Barry Gaudette, a fiber and hair analysis expert from the Royal Canadian Mounted Police, visited Atlanta from December 7 to December 18, 1981. He conducted fiber and hair comparisons in three of the cases then under investigation, including those of Jimmy Ray Payne and Nathaniel Cater, for whose murders the appellant was ultimately convicted. He did not prepare written reports of his findings and no information regarding his research was provided to the defendant. Trial began on December 28, 1981, and during the trial Gaudette testified from personal notes as to the results of the fiber tests he had conducted. Defendant objected on the *827 ground that he had not been provided a copy of these test results pursuant to his request under OCGA 17-7-211 (Code Ann. 27-1303). Unlike *Law*, there is no indication here that Gaudette was rushed or hurried in completing his tests or preparations for testifying. The only similarity with *Law* is that the state and its expert witness have openly and clearly been allowed to flaunt the provisions and purpose of OCGA 17-7-211 (Code Ann. 27-1303). We are now faced with exactly the evil I anticipated *Law* would spawn, and the majority continues to encourage rather than eradicate this evil.

Appellant also contends that the personal notes and test results compiled by an expert witness should be included within the scope of "scientific evidence" under OCGA 17-7-211 (Code Ann. 27-1303). The majority disagrees and follows decisions of the Court of Appeals which conclude that a witness' work product is not discoverable. We need not overrule those cases, but we do need to add that where only notes and test results exist and are used to support and refresh the expert's memory during trial, then those materials should be discoverable under OCGA 17-7-211 (Code Ann. 27-1303) to the extent that they embody the results of scientific tests. By allowing an expert to forgo delivery of a full written report and to later testify orally where, as here, he had ample time to prepare such a written report and conducted tests too complex to remember unaided, we permit ever more egregious injustice and violation of the intent of OCGA 17-7-211 (Code Ann. 27-

1303), which is to put into the defendant's hands these reports with sufficient time before trial to enable him to check and challenge their content.

4. I turn now to appellant's claim that he was denied effective assistance of counsel. Although the majority declines to review this issue, I find no justifiable reason for doing so where, as in this case, the ineffectiveness of trial counsel is plain and egregious and can be discerned from the record and transcripts.

This court has interpreted effectiveness of counsel to mean "not errorless counsel, and not counsel judged ineffective by hindsight, but counsel reasonably likely to render and rendering reasonably effective assistance." *Johnson v. Zant*, 249 Ga. 812 (295 SE2d 63) (1982) (quoting *MacKenna v. Ellis*, 280 F2d 592, 599 (5th Cir. 1960)); *Pitts v. Glass*, 231 Ga. 638, 639 (203 SE2d 515) (1974). Moreover, we have noted that when inadequate representation is alleged the crucial inquiries are ordinarily whether the defendant had a defense which was not presented; whether trial counsel adequately consulted with the accused and adequately investigated the facts and the law; and whether the omissions charged to trial counsel resulted from inadequate preparation rather than from "unwise choices of trial tactics and strategy." *Hudson v. State*, 250 Ga. 479 (8) (299 SE2d 531) (1983). The amount of investigation that is adequate "will necessarily depend upon a variety of factors including the number of issues in the case, the relative complexity of those issues, the strength of the government's case, and the overall strategy of trial counsel." *Washington v. Strickland*, 693 F2d 1243, 1251 (5th Cir. 1982), cert. granted, 51 USLW 3871 (6/6/83). However, trial counsel must "conduct appropriate investigations, both factual and legal, to determine what matters of defense can be developed." *Id.* at 1254.

At the outset I note that Williams had the services of many attorneys during the course of his trial, and that they were confronted with many difficult tasks in investigating his case and preparing defenses for trial, including deciding which of hundreds of witnesses to interview and then interviewing them, and investigating and examining the voluminous, complicated, and novel fiber evidence accumulated by the state. Nevertheless, for the reasons which follow I find that Williams' defense counsel rendered ineffective assistance because of their failure to adequately investigate the law and facts concerning several lines of defense which they pursued at trial.

Perhaps most importantly, although Williams was entitled, pursuant to *Sabel v. State*, 248 Ga. 10 (6) (282 SE2d 61) (1981), to have an expert of his own choosing examine the critical fiber evidence used by the state to implicate him in the two charged and ten uncharged murders, many actions of defense counsel emasculated Williams' rights under *Sabel* and

rendered inadequate the defense's investigation of the fiber evidence and its development of a defense thereto.

First, Williams' attorneys were dilatory in choosing an expert to examine this evidence. Despite knowledge of the important role an expert would play in Williams' defense, and despite several promptings by the trial court in September and the early part of October of 1981 to quickly obtain and submit an expert for its approval, defense counsel did not submit their chosen expert, a Dr. Morton, for the trial court's approval until November 12, 1981. Second, after choosing Dr. Morton, defense counsel failed to coordinate his schedule with that of the crime lab, with the result being that on several occasions he came to Atlanta from California and was unable to gain access to the crime lab and the evidence and equipment therein. Third, despite the pre-trial knowledge that the state intended to introduce evidence concerning at least a portion of the other twenty-seven murders on the task force list, the defense did not request, either in its August 14, 1981 Sabel motion or at any other time prior to trial, that the court require the state to disclose which of *829 these crimes it intended to introduce and then allow the defense access to the fiber evidence pertaining to each of them. (The trial court's November 12, 1981 Sabel order only granted the defense access to the fiber evidence pertaining to the Payne and Cater cases.) Furthermore, after the state, on January 22, 1982, pinpointed the other crimes it wished to introduce, and after the court, on January 25, 1982, granted the state's motion to be allowed to present evidence on these murders, Williams' attorneys did not move for a continuance to enable Dr. Morton to examine the fiber evidence in those cases. In fact, it was not until February 1, 1982, when the defense made a motion to suppress evidence relating to the ten independent murders, that the defense expressed dissatisfaction with the trial court's failure to authorize its expert to inspect the fiber evidence pertaining to the ten extrinsic murders. Additionally, the defense failed to take advantage of two opportunities given to it by the state to examine the fiber evidence in the independent offenses.

Although all of the above omissions rendered trial counsel's preparation of an adequate defense to the state's fiber evidence practically hopeless, the fatal blow to defense counsel's effectiveness in this area was their failure to have Dr. Morton testify at trial. Morton apparently left Atlanta towards the end of January or early February 1982 and refused to return, thus forcing the defense to obtain a second expert in the twilight of the trial. The result was that this expert spent less than five hours examining the state's voluminous collection of fiber evidence. Regardless of the reasons why Morton refused to return to Atlanta, defense counsel's failure to take the necessary measures to assure that he would testify is inexcusable. Morton was the only expert Williams had who had had enough

time to review the fiber evidence, the linchpin of the state's case, and help prepare a defense thereto.

Related to the above ineffectiveness of defense counsel was their failure to challenge the scientific reliability of the principles and techniques used by the state's experts in conducting their fiber comparisons. See Division 2 of this dissent. This challenge was crucial to an adequate defense to the fiber evidence, and its omission demonstrates inadequate preparation and investigation by defense counsel and is a significant example of counsel's ineffectiveness.

Another glaring failure by defense counsel to investigate the law and facts concerns their failure to adequately raise the issue of or invoke a ruling by the trial court on the state's failure to advise Wayne Williams of his Miranda rights on May 22, 1981, a night on which as many as ten police officers and FBI agents interrogated Williams for approximately ninety minutes after stopping him on Interstate 285.

In addition to the above failures by defense counsel to adequately *830 investigate the law and facts relative to certain defenses, the trial transcript reveals that in many instances defense counsel waived the right to object to or failed to object to certain prejudicial evidence introduced at trial and to certain improper conduct which occurred at trial. These omissions include: 1) the failure to object to the highly questionable and prejudicial use by a state's witness of the calculus of compound probabilities to explain the probative force of certain fiber evidence (this includes the use of hearsay by the state's expert, to which defense counsel did not interpose an objection concerning Williams' right of confrontation) (see Division 1 of the majority opinion and Division 2 of this dissent); 2) the failure to object to a prosecutor's use of these probabilities, supplemented with mathematical odds based upon his own calculations, during closing argument; 3) the waiver of the defense's presence on certain occasions when the trial court talked with the jury (see Division 11 of the majority opinion); 4) the failure to object to improper closing argument (see Division 14 of the majority opinion); 5) the waiver of the right to object to Dr. Zaki's references to other crimes (see Division 4 of the majority opinion); 6) the waiver of the right to object to the prosecution's question to Tilbert Boynham concerning whether he had been given a lie detector test (see Division 15 of the majority opinion); and 7) the defense's failure to object to the trial court's failure to preserve for appellate review the Brady material it inspected. It is clear that these omissions, when considered together, were prejudicial to Wayne Williams.

For the above reasons, I find that Wayne Williams' attorneys, due to inadequate preparation and inadequate investigation into the law and facts of their client's case,

rendered him ineffective assistance of counsel.[8] I would therefore reverse his convictions and remand the case for a new trial.

NOTES

[1] We are not here dealing with the question of whether the requirements of OCGA 17-7-211 (Code Ann. 27-1303) are violated if a district attorney knows of the results of a scientific experiment and specifically instructs the examiner or expert not to reduce his writings to the form of a written report. See *Law v. State*, supra, dissenting opinion of Weltner, J.

[2] We note that the defendant was in fact given considerable latitude in voir dire. For example, after the exchange quoted above, he was allowed to ask, "Would you tell me that if you were selected as a juror in this case, you would sit and wait and hear the prosecution's evidence and wait for the defendant's evidence to come in?" We find no merit in the argument that voir dire was impermissibly restricted.

[1] The meaning of the majority's references to "pattern" as a purpose for which the other crimes evidence was admissible is unclear. "Pattern" is not a recognized exception to the general rule excluding evidence of other crimes. *McCormick*, supra, 190, pp. 447-451. Although the trial judge in this case ostensibly admitted the evidence of 10 other crimes to show Williams' "plan, scheme, pattern, bent of mind, and identity," plan, scheme, and pattern are properly thought of as components of the traditional "plan or scheme" exception, which is relevant to the ultimate issue of identity. See *McCormick*, supra, at 448. "Bent of mind" has no application here, since intent was not a contested issue at trial. I conclude that identity (and particularly, modus operandi) was the only relevant purpose for introduction of this evidence. Identity may be inferred from other crimes evidence showing a plan or scheme, motive, or modus operandi. See pp. 4-6, *infra*; *McCormick*, supra, 190, pp. 448-451; *Wright & Graham*, *Federal Practice and Procedure* 5246, pp. 511-512 (1978); *United States v. Myers*, 550 F2d 1036, 1045 (5th Cir. 1977).

[2] Webster's Third New International Dictionary (1971) defines "modus operandi" as "a distinct pattern or method of procedure thought to be characteristic of an individual criminal and habitually followed by him . . ." *Id.* at 1453.

[3] The majority opinion again fails to articulate a clear standard in this case, stating only that the accused's perpetration of the extrinsic offenses need not be proven "beyond a reasonable doubt." I find it interesting that this court, which sua sponte created a "clear and convincing" standard of appellate review in child custody cases, see *Blackburn v.*

Blackburn, 249 Ga. 689 (292 SE2d 821) (1982), does not recognize the need for a clear standard of proof to guide judges and juries in criminal trials involving extrinsic offenses.

[4] Studies have shown that introduction of other crimes evidence has a decided impact on the outcome of criminal trials. See Note, 76 Colum. L. Rev. 327, 343 n. 95 (1976) ("[W]hen a defendant's criminal record is known and the prosecution's case has contradictions, the defendant's chances of acquittal are 38%, compared to 65% otherwise.")

[5] The record indicates that defense counsel failed to object at trial to the state's fiber evidence on the grounds of scientific unreliability. Despite this fact, the majority opinion appears to address the merits of the reliability issue. Therefore I feel compelled to state my views on this issue.

[6] The experts declined to attribute this level of significance to the fibers found on Joseph Bell, apparently because so few were found which compared favorably with the Williams fibers.

[7] This fact, critical to the state's "uniqueness" argument, was never conclusively established and was indeed questionable in light of testimony (based on Wellman sales records) that Wellman fiber was sold to a number of Georgia and southeastern manufacturers during the period in question.

[8] I believe that, in light of the compulsory legal education requirements adopted by this court and effective January 1, 1984, this court should establish a procedure whereby any attorney found to be ineffective would automatically be referred to the State Disciplinary Board for investigation and appropriate sanctions.

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